A Correlation-Aware Graph Transformer Framework for Joint Regional Load and Operational Stress Forecasting in Bangladesh

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***Abstract*—Day-ahead planning in developing-economy national grids requires synchronised visibility over regional electricity demand and system-wide operational pressure, yet load prediction and stress monitoring are often handled separately. On the Bangladesh national power system, daily records from nine administrative divisions jointly report evening-peak load with limitation, reserve, and shedding indicators, making coupled forecasting necessary. This study examines whether a spatio-temporal model can jointly predict next-day regional demand and a graph-level Operational Stress Index (OSI) from shared historical inputs under chronological evaluation. We implement the Correlation-Aware Multi-Task Forecasting Framework as a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT) that encodes a seven-day lookback of division-level and national context with a train-estimated correlation graph, then decodes regional demand and a bounded system-wide stress scalar through shared spatio-temporal encoding and task-specific heads, using a repaired multi-task loss to stabilise joint learning. On a frozen held-out test year, evaluation includes classical, recurrent, and graph-convolution benchmarks, graph and architecture ablations, regional error analysis, and multi-method post-hoc attribution. Within the Bangladesh setting, correlation-graph PF-STGT achieves the strongest joint demand-and-stress performance in the internal suite, with statistically supported demand improvement over the historical hybrid reference and comparatively informative OSI outputs relative to stress-capable baselines. Ablation favours correlation adjacency over geographical or hybrid priors, documents a demand-only versus joint-stress trade-off, and attribution analysis highlights limitation and calendar coalitions with partial cross-method agreement. The paper presents a reproducible national case study of graph-transformer multi-task forecasting scoped to offline evaluation rather than universal superiority or demonstrated deployment outcomes.**

***Index Terms*—Multi-task learning, Spatio-temporal forecasting, Operational stress index, Load forecasting, Correlation graph, Smart grid, Explainability, Bangladesh power grid**

I. Introduction

National grid operators must anticipate regional load and assess system pressure on a day-ahead horizon to coordinate generation dispatch, reserve planning, and constraint management. In developing economies, supply limitations, infrastructure stress, and load shedding often coincide with rapid demand growth, so planners need forecasts that show both where load will concentrate across regions and how severely operational constraints may bind system-wide. Many South Asian systems report daily division-level balances, national generation scalars, and limitation indicators that nevertheless encode stress when reserves tighten or infrastructure constraints bind. On the Bangladesh national power system, operational data are reported across nine administrative divisions over a multi-year timeline, with regional evening-peak demand and limitation dynamics co-visible in each daily record [1]. Effective planning therefore requires spatially disaggregated, temporally aligned forecasts that support review when stress indicators and load trajectories must be judged together.

Despite these data, forecasting practice often treats load prediction and stress monitoring separately. Classical and ensemble models [2]–[4] forecast demand without explicit inter-division graph structure, while shallow recurrent or graph-convolution architectures [5], [6] may fail to combine long-range temporal context with informative multi-node representations on national footprints. Graph-based methods [7]–[9] from residential or microgrid settings often rely on geographical adjacency that may not capture data-driven co-movement among regional loads. Operational stress—from reserve shortfall, limitation pressure, and shedding intensity—is seldom forecast jointly with regional demand; shedding studies [10], [11] typically address control and optimisation rather than day-ahead graded stress indices. Explainability, where present [12], [13], is rarely integrated with multi-node graph-temporal models that emit coupled load and stress outputs. Operational records thus offer synchronised regional load and constraint context, whereas many pipelines deliver disconnected single-task predictions with limited spatial structure and transparency.

We address this mismatch by formulating day-ahead forecasting as a supervised multi-task problem [14] over nine spatial units. The primary task is next-day regional evening-peak demand; the secondary task is a graph-level Operational Stress Index (OSI) summarising system-wide pressure from shedding intensity, generation reserve margin, and operational limitation components normalised under a leakage-safe training policy. OSI is a bounded composite score rather than a sparse per-division shedding label, preserving graded stress magnitude when reserve or limitation pressure is elevated even without shedding. Demand and OSI are operationally related yet statistically non-redundant: demand describes expected load by region, whereas OSI encodes constraint-related pressure not fully determined by demand alone. The research problem is whether a single spatio-temporal framework can learn coupled regional demand and operational stress forecasts from historical Bangladesh grid records under chronological evaluation, and which design choices—graph prior, architecture, task weighting, and post-hoc attribution—are empirically justified on this case.

A structured literature review identifies further gaps motivating this work. Bangladesh division-level smart-grid daily data remain rare in peer-reviewed spatio-temporal load and shedding studies; developing-economy national grids are under-represented relative to US and European benchmarks. Multi-node demand forecasting is seldom coupled with operational-stress targets in one multi-task framework, and shedding research rarely targets forecast-centric graded daily stress. Only a minority of recent work combines graph-based spatial coupling with transformer-style temporal modelling [15]–[17] for multi-node national forecasting, and none address the Bangladesh shedding and limitation context directly. Explainable artificial intelligence appears in few graph-temporal energy studies [12], [18] and is seldom reported with multi-method discordance for coupled demand and stress outputs. Weak documentation of chronological splits, baselines, and ablations in part of the conference literature further complicates fair comparison. These gaps motivate a reproducible, evidence-locked study that couples graph-temporal learning, multi-task stress forecasting, and operator-oriented attribution within one frozen evaluation programme.

This paper proposes the Correlation-Aware Multi-Task Forecasting Framework, implemented as a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT). The framework ingests a seven-day lookback of node-level and national context features with a train-estimated correlation graph encoding inter-regional demand coupling from training data only. A parallel dual-branch encoder combines graph-aware spatial attention with temporal transformer capacity [7], [17]; learned fusion feeds task-specific heads that decode regional megawatt demand and a bounded system-wide OSI scalar for the next operating day. Graph construction compares geographical, hybrid, and correlation-only priors; the final configuration adopts correlation-thresholded adjacency following empirical comparison. Multi-task training uses scale-aware joint loss with explicit stress weighting [14], [19] so shared representations retain limitation-relevant structure without collapsing to demand-only learning. Post-hoc attribution applies grouped integrated gradients, permutation importance, and spatial and temporal attention export [2], [18], [20], together with stratified case studies on the frozen checkpoint, with method disagreement reported explicitly. Evaluation on chronologically partitioned data includes classical, recurrent, and graph-convolution benchmarks [2], [5], [21], configuration ablations, architecture simplification, error geography, and explainability analysis; protocol details appear in later sections.

The principal contributions are as follows. First, a Correlation-Aware Multi-Task Forecasting Framework (configuration S2) for joint day-ahead regional demand and graph-level OSI prediction from a single spatio-temporal graph-transformer on Bangladesh national grid data. Second, evidence-driven architecture selection that replaces the initial hybrid geographical–correlation graph with correlation-only adjacency while retaining full temporal encoder capacity where ablation shows it is consequential. Third, a repaired multi-task training protocol with explicit demand–stress loss balancing under the frozen W20 specification. Fourth, an empirical evaluation programme—external benchmarks, component ablations, inferential testing on primary demand MAE with multiple-comparison correction, and metric verification for classical baselines—supporting reproducible comparison within the frozen suite. Fifth, integrated explainability analysis with multi-method attribution artefacts for both tasks, including spatial and temporal views and comparison against OSI component decomposition on audited case studies. These contributions rest on offline evaluation on held-out test data and are positioned as planning and transparency evidence rather than proof of deployment outcomes or universal architectural superiority.

The remainder of this paper is organised as follows. Section II reviews related work; Section III presents the methodology; Section IV describes the experimental setup; Section V reports results; Section VI discusses findings and limitations; and Section VII concludes. Supplementary materials are provided in Appendix A.

II. Related Work

We survey prior research along four themes—short-horizon load forecasting, graph-based spatio-temporal modelling, multi-task learning in energy systems, and explainable artificial intelligence—drawing on a structured review of fifty-five studies (2023–2026) from the Bangladesh smart-grid programme. The focus is on methodologies, assumptions, and unresolved challenges that bound the Correlation-Aware Multi-Task Forecasting Framework (PF-STGT), not on enumerating individual publications.

A. Load Forecasting

Short-horizon electricity load forecasting remains the largest cluster in the reviewed literature. Methods range from classical regressors to deep recurrent, convolutional, and transformer architectures applied mainly to scalar or multivariate load series. Much of this work targets residential or AMI consumption at sub-daily resolution, assuming dense sampling rather than administratively defined regional nodes on a national footprint. Ensemble trees and gradient boosting remain strong tabular baselines when spatial structure is implicit in feature engineering [2]–[4].

Deep learning studies emphasise temporal pattern extraction through stacked recurrent encoders, Autoformer- and transformer-style models [5], [22]–[24], and multi-encoder designs that fuse exogenous embeddings with load history. Common assumptions include stationary seasonality, long uninterrupted series, and horizons aligned to market cadences that may differ from daily national operational reporting. Developing-economy national grids with division-level evening-peak reporting and co-visible limitation indicators are effectively absent from the collected corpus.

A parallel stream addresses load shedding, under-frequency response, and emergency control rather than ex-ante day-ahead forecasting [10], [11]. These studies optimise shedding sequences, preserve stability, or manage islanded microgrids using simulation-based or event-centric datasets. Forecasting graded daily stress for planning is related to shedding research but distinct from optimising control actions at disturbance timescales.

Across this cluster, external-validity reporting is often weak—South Asian division-level case studies are rare, conference abstracts can be metadata-sparse, and evaluation metrics are sometimes mixed without clear documentation. Open challenges include adapting AMI-centric pipelines to daily national timelines, modelling regional heterogeneity where one division dominates load, and coupling load forecasts with system-pressure indicators derived from limitation dynamics.

B. Graph-based Spatio-Temporal Forecasting

Graph neural networks and graph-attention architectures [7], [25] construct nodes from geographic areas, charging stations, or feeders, with edges defined by administrative proximity, infrastructure topology, or correlation in historical series. Spatio-temporal graph convolutional networks propagate information across fixed neighbourhoods before or after temporal encoding [8], [9], [26].

Transformer-based spatio-temporal forecasting stacks graph attention with temporal transformers or applies dual-channel spatial–temporal decoders [15]–[17], typically on US or European residential benchmarks with spatial semantics unlike a nine-division national grid. Geographic adjacency is a common inductive bias; data-driven correlation graphs appear less often as the sole spatial prior, and hybrid geographic–correlation constructions are seldom ablated against correlation-only alternatives on developing-economy records.

Dataset assumptions diverge from the Bangladesh setting: node counts, sub-hourly resolution, and rare co-forecasting of stress with regional demand in the same graph-transformer pipeline. Shallow graph-convolution baselines [6], [21] test whether graph propagation alone suffices when temporal encoding is limited. Reporting of graph-construction splits is often insufficient, explainability of learned spatial weights remains sparse, and evidence is limited on when geographical priors help versus add noise on strongly correlated national demand series. Selecting graph priors when inter-regional coupling reflects shared national growth rather than border adjacency, determining whether transformer capacity is conditional on graph topology, and extending graph-temporal models to division-level nodes with synchronised limitation covariates remain open problems. None of the high-relevance collected studies address Bangladesh shedding context or daily OSI-style stress targets.

C. Multi-Task Learning

Multi-task learning in energy systems [14] occupies a narrow band relative to single-task load forecasting and shedding control. Reported formulations couple related energy vectors or hierarchical horizons within a single sector [19], [26]. Explicit coupling of continuous multi-node regional demand with a daily operational stress index derived from shedding intensity, reserve margin, and limitation pressure is not represented among collected comparators.

Task isolation remains dominant: demand papers treat load as the sole target; shedding papers optimise control; operational-stress studies focus on transmission assets rather than forecastable daily stress scalars. Auxiliary targets, where present, are not paired with graph-level spatial encoders on national regional footprints. The literature offers limited guidance on balancing megawatt-scale robust regression against bounded stress targets or on repairing stress-head collapse when naive multi-task weighting yields uninformative auxiliary outputs.

Scarce ablation of single-task versus multi-task ceilings, infrequent reporting of Pareto trade-offs when auxiliary tasks marginally degrade primary metrics, and absence of developing-economy national case studies with documented chronological splits further limit transferability. Open questions include when multi-task learning is preferable to separate pipelines, how to calibrate task weights under composite-target semantics, and whether shared graph-temporal encoders transfer stress-relevant structure beyond single-task models with global limitation covariates.

D. Explainable AI

Explainability for load and smart-grid forecasting is sparsely represented [12]. Where it appears, it clusters into SHAP attributions for control or shedding policies [11], [18], interpretable ML frameworks for decentralised prediction [13], and review articles cataloguing feature-importance practices for scalar load series without graph-temporal multi-task outputs.

Attribution is often treated as a supplement to control validation rather than operator-facing diagnostics for coupled regional demand and national stress forecasts. Graph-temporal models rarely export spatial attention, coalition-grouped gradients, and dual-path comparison against engineered stress components in one evaluation programme. Review-level work [12] acknowledges that explainability methods can disagree on feature rankings, yet few studies document that disagreement for multi-node graph transformers.

Attributions are frequently computed on small validation samples without operator studies, post-hoc sensitivity is sometimes conflated with physical causation, and stress and demand explanations are seldom jointly audited. Standardising multi-method attribution for dual-task dashboards, aligning explanations with engineered stress components, and extending microgrid-oriented SHAP workflows to national graph models with eleven feature coalitions remain unresolved.

E. Summary and Research Gap

The reviewed literature offers capable methods within each thematic silo but seldom integrates them for the problem addressed here. Four cross-cutting gaps emerge.

First, geographic validity is limited: no collected peer-reviewed spatio-temporal study uses Bangladesh division-level smart-grid daily data [1], and developing-economy national grids remain under-represented. Second, task formulation remains fragmented: continuous regional demand is rarely coupled with operational-stress targets in one multi-task graph-temporal model, while shedding research predominantly targets control rather than day-ahead planning visibility. Third, methodological integration is incomplete: only a minority combine graph-based spatial coupling with transformer-style temporal modelling, and few co-design limitation-aware covariates with spatio-temporal architectures or document chronological evaluation with ablations and corrected inferential testing. Fourth, explainability is rarely embedded in coupled demand-and-stress graph forecasts with explicit multi-method reporting.

These gaps motivate PF-STGT as a targeted response rather than a claim of universal superiority. The framework predicts joint next-day regional evening-peak demand and graph-level OSI on Bangladesh operational records under leakage-safe chronological evaluation, using a train-estimated correlation graph as an empirically testable spatial prior, parallel-fusion graph-aware and temporal transformer encoding, and a repaired multi-task loss that stabilises stress learning. Post-hoc attribution—coalition-level gradients, permutation comparisons, attention export, and case-stratified dual-path stress review—is treated as a reporting obligation. The present study sits at the intersection of Bangladesh division-level analytics, multi-task demand-and-stress coupling, graph-temporal modelling with documented graph-prior ablation, and explainability on frozen multi-task outputs. Methodology and empirical evaluation follow in subsequent sections.

III. Methodology

A. Problem Definition and Task Formulation

Operators of modern power systems must balance generation dispatch, reserve allocation, and network security under growing load and supply variability. Historical operational data motivate spatio-temporal frameworks that exploit regional load coupling and system-wide context. Operators also need visibility into operational stress from reserve shortfall, fuel and infrastructure limitations, and load-shedding pressure. This study addresses both requirements on the Bangladesh national power system; dataset properties appear in Section III-B.

We formulate day-ahead forecasting as a supervised multi-task learning problem [14] over a set of spatial units indexed by graph nodes. Let $\mathcal{V} = \{1, \dots, N\}$ denote the node set, with N regional divisions. Time is indexed in daily steps t. At each forecast origin t, the model receives a lookback window of length T consecutive observed days and predicts targets at horizon H. Inputs and outputs are defined over the window $[t - T + 1, t]$ for observations and at $t + H$ for supervision. The batch dimension is omitted where clear from context.

Inputs comprise two coupled tensors. Regional node features are denoted $\mathbf{X}^{\mathrm{node}} \in \mathbb{R}^{T \times N \times F_n}$, where F_n is the number of node-level features per region per day, encoding division-level historical demand signals, engineered lag and rolling statistics, and related regional observables. National and system-level context is represented by $\mathbf{X}^{\mathrm{global}} \in \mathbb{R}^{T \times F_g}$, where F_g is the number of global features per day, including calendar indicators, grid aggregates, operational limitation variables, and national generation scalars. Together, $\mathbf{X}^{\mathrm{node}}$ and $\mathbf{X}^{\mathrm{global}}$ represent all information available at or before day t from which forecasts are produced.

Task 1 produces a regional demand vector $\hat{\mathbf{y}}^{\mathrm{demand}} = (\hat{D}_1, \dots, \hat{D}_N)^\top \in \mathbb{R}^N$, with each $\hat{D}_r$ expressed in megawatts (MW). Task 2 produces a graph-level operational stress scalar $\hat{y}^{\mathrm{osi}} \in [0, 1]$. The demand output assigns one next-day prediction per node; the stress output is a single system-wide score rather than a per-region vector.

For Task 1, each node $r \in \mathcal{V}$ predicts next-day regional evening-peak demand $D_r(t + H)$ in MW. Node-level targets capture distinct load scales and spatially coupled dynamics that a national aggregate cannot represent. Regional supply is excluded because prior analysis documented strong demand–supply collinearity; division-level shedding is not regressed directly because it is sparse and zero-inflated at the regional level. Shedding intensity enters instead through the composite stress target below.

Task 2 predicts the Operational Stress Index (OSI), a bounded system-health score integrating shedding severity, generation reserve margin, and operational limitation pressure. OSI combines three components normalised using train-split bounds only:

$$c_1 = \frac{L_{\text{total}}}{D_{\text{total}}}, \qquad c_2 = 1 - \frac{GR}{\text{Highest\_Generation}}, \qquad c_3 = \frac{\text{TOL}}{\text{Highest\_Generation}}, \tag{1}$$

$$\text{OSI} = \text{mean}\big(\text{minmax}_{\text{train}}(c_1), \text{minmax}_{\text{train}}(c_2), \text{minmax}_{\text{train}}(c_3)\big), \tag{2}$$

where L_{total} is total regional shedding (MW), D_{total} is total regional demand, GR is generation reserve, and TOL is total operational limitation. Continuous OSI regression preserves graded stress magnitude, including reserve- and limitation-driven pressure without a shedding event. The supervised target is $\text{OSI}(t + H)$.

Regional demand and OSI are learned jointly because the targets are operationally related yet statistically non-redundant. Demand forecasts describe expected load levels across spatial units, whereas OSI summarises constraint-related system pressure through components c_1–c_3 that are not fully encoded by demand alone. Joint estimation provides synchronised day-ahead visibility over load and stress from shared calendar and system context. A leakage policy applies to stress supervision: when predicting $\text{OSI}(t + H)$, same-day $\text{OSI}(t)$ is excluded from the input feature set; only information observable at or before t may be used.

Both tasks use a single-step horizon $H = 1$ day. At origin t, the model predicts $\{D_r(t+1)\}_{r\in\mathcal{V}}$ and $\text{OSI}(t+1)$ from inputs through day t, aligning with the daily reporting cadence, regional demand structure, and lag-based feature design. Longer horizons are reserved for future work.

The multi-task objective combines Task 1 and Task 2 losses. Let $\mathcal{L}_{\text{demand}}$ denote a robust regression loss on the MW scale—implemented as Huber loss [27] averaged over the N regional demand targets—and let $\mathcal{L}_{\text{osi}} = \ell_{\text{MSE}}(\hat{y}^{\text{osi}}, \text{OSI}(t + H))$ denote mean squared error on the bounded stress target. The joint training objective is

$$\mathcal{L} = \mathcal{L}_{\text{demand}} + \lambda_2\, \mathcal{L}_{\text{osi}},$$

where $\lambda_2 > 0$ weights stress learning against demand fit. Task-weight calibration and optimisation details follow in Sections III-G and III-H.

B. Dataset Description

We apply the multi-task formulation to historical operational records from the Bangladesh national power grid using the publicly available Bangladesh Smart Grid dataset (Mendeley Data, Version 5) [1]. The daily operational timeline spans several years of system and regional reporting; key properties are summarised in Table I.

Data source and granularity. The underlying archive provides one record per calendar day. Each record aggregates national generation and demand indicators together with division-level operational signals. The timeline spans 21 November 2019 to 30 December 2024, yielding 1,850 unique daily observations over a 1,866-day calendar span. Sampling is predominantly at one-day intervals; a small number of calendar gaps are present in the series and are accounted for in later modelling stages. All 45 recorded fields are complete, with no missing entries in the audited source timeline. Table I lists the domain, regional coverage, and structural properties used in this study.

Spatial coverage. The spatial units in $\mathcal{V}$ correspond to $N = 9$ administrative divisions of Bangladesh: Barishal, Chattogram, Cumilla, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, and Sylhet. This partitioning matches the regional reporting structure reflected in the source data and supports node-level demand forecasting across geographically distinct load centres.

Recorded variables. The daily records combine calendar metadata, national generation and demand scalars, exogenous operational constraints, and regional power-balance signals. Calendar fields include the date, day of week, month, year, and holiday indicators. National-level quantities document evening-peak and day-peak generation, minimum and forecast generation, and demand measured at generation and substation endpoints. Operational constraint fields capture gas and liquid-fuel limitations, coal supply limitations, low water levels at Kaptai Lake, and capacity under plant shutdown or maintenance. A national temperature anomaly proxy for Dhaka is also recorded. At the regional level, each division is associated with three coupled signals: evening-peak demand, supply, and load (unmet demand / shedding) in megawatts. In total, the source timeline contains 42 numeric operational variables and three non-numeric calendar descriptors alongside the date index.

Target variables. Two supervised targets are derived from the recorded timeline in alignment with Section III-A. For Task 1, the primary target is next-day regional evening-peak demand $D_r(t + H)$ for each node $r \in \mathcal{V}$, taken from the division-level demand fields. For Task 2, the Operational Stress Index is a graph-level score in $[0, 1]$ constructed from system-wide shedding intensity, generation reserve margin relative to highest available generation, and aggregate operational limitation pressure—the inputs to components c_1–c_3 defined in Section III-A. Regional supply and division-level shedding are retained in the source record because they inform system context and stress composition, but only demand and OSI serve as forecasting targets under the frozen task definition.

Dataset characteristics. The timeline suits joint demand–stress forecasting: daily cadence matches $H = 1$; nine divisions provide a natural node set; demand, shedding, reserve, and limitation co-exist in each record for composite stress construction; and multi-year coverage spans full seasonal cycles, holidays, and supply-constraint periods.

Prior data review documented strong demand–supply coupling, motivating demand-only Task 1 targets, and sparse regional load distributions, motivating OSI rather than per-node shedding regression. Inter-regional demand correlation supports graph-based spatio-temporal modelling in subsequent subsections.

C. Data Preprocessing

The operational timeline described in Section III-B is prepared for downstream modelling through a two-stage preprocessing pipeline that precedes feature engineering. The first stage performs conservative scientific cleaning to validate and correct the source record while preserving genuine operational events. The second stage converts the cleaned timeline into a model-ready representation through chronological ordering, categorical encoding, and numeric standardisation, with all learnable transformations restricted to training data to prevent temporal leakage. No lag features, rolling statistics, graph structures, or supervised targets are created in this stage.

Data validation and integrity assessment. Preprocessing begins with a read-only audit of the published raw archive to establish record counts, field types, and temporal structure without altering source values. The audit confirmed 1,850 daily rows and 45 fields, with 42 numeric operational measurements and three non-numeric calendar descriptors indexed by date. Missing-value assessment across all fields returned zero absent cells. Duplicate inspection found no repeated rows and no duplicate dates. Negative measurements and implausible temperature values were absent. These checks establish that the archive is complete at the cell level and suitable for time-ordered forecasting, while deferring any modification of the raw file to the cleaning stage.

Scientific data cleaning. The cleaning stage ingests the published archive and produces a cleaned intermediate dataset of dimensions 1,850 $\times$ 45. Cleaning follows a conservative rule: only impossible, corrupted, duplicated, or invalid records are corrected or removed; statistically rare but operationally genuine events are retained. No rows were removed. The date field was validated and converted to a standard temporal representation without changing calendar values. Timestamp verification confirmed that all dates are parseable, unique, and strictly monotonic increasing over the range 2019-11-21 to 2024-12-30. Chronological consistency checks verified agreement between the date index and the recorded year and month fields in every row. Six inconsistent day-of-week labels were corrected deterministically from the authoritative date index; no numeric measurements were altered.

Calendar continuity was assessed against a daily index. Seventeen missing calendar days were documented in the series; these gaps were not imputed or synthetically filled during cleaning, so only observed daily records are retained. This policy preserves the authentic reporting cadence and avoids introducing artificial observations that would distort lag-based forecasting. Physical-consistency review identified regional accounting mismatches and generation-ordering discrepancies in a subset of rows; these patterns were classified as reporting or metering differences rather than impossible values and were preserved as informative operational signals. Row and column counts were unchanged between input and output, confirming that cleaning altered only invalid metadata rather than measurement content.

Model-ready preprocessing. The cleaned intermediate timeline serves as the sole input to the model-preparation stage. Records are sorted in ascending chronological order to enforce temporal ordering required for causal forecasting. Three categorical fields—day of week, holiday name, and holiday category—are one-hot encoded using categories observed in the training partition only, with unseen validation or test categories assigned zero contribution. This encoding supplies a fixed numeric representation of calendar and holiday context without manual category collapsing.

Forty-one numeric operational fields, comprising national generation and demand scalars, limitation indicators, temperature, calendar ordinals (year and month), and all regional demand, supply, and load measurements, are standardised by z-score scaling with parameters learned exclusively from the training partition. Validation and test partitions are transformed using the frozen training statistics only. Standardisation places heterogeneous MW-scale quantities on a common scale for subsequent learning while preventing held-out data from influencing the fitted parameters. The date index is retained as an unscaled temporal key. Only existing source fields are transformed; no new predictive attributes are introduced at this stage.

Leakage prevention. Temporal leakage is controlled at two levels during preprocessing. First, chronological ordering is preserved throughout cleaning and transformation so that no future observation precedes a past one in the stored timeline. Second, all learnable preprocessing parameters—the one-hot category sets and scaling means and standard deviations—are estimated on the training partition only; validation and test data receive transformation with frozen training parameters. Cleaning and model-preparation stages remain separable, so encoding and scaling cannot retroactively alter the cleaned record. Same-day stress indices and other supervision-derived quantities are not injected during this stage; the leakage policy for OSI inputs defined in Section III-A is enforced in subsequent feature construction.

Preprocessing outputs. The pipeline delivers a validated cleaned dataset, chronologically partitioned encoded records, and a serialised transformation pipeline freezing encoding and scaling rules for downstream feature engineering, graph construction, and multi-task training.

D. Feature Engineering

The leakage-aware preprocessing outputs described in Section III-C are extended into a structured feature set that supplies the spatio-temporal inputs defined in Section III-A. Feature engineering transforms the cleaned operational timeline and its encoded baseline representation into derived predictors that capture autoregressive load dynamics, calendar structure, grid-wide stress context, and regional heterogeneity. The design prioritises causal observability: every engineered quantity at day t is computable from information available at or before t, with no reliance on future rows or held-out partition statistics. In total, 65 engineered features are constructed and merged with the preprocessed baseline by date. Raw source measurements and their preprocessed encodings are distinguished throughout from quantities constructed by deterministic transformation rules.

Design objectives. The feature programme serves three methodological goals aligned with the joint demand–stress formulation. First, it encodes short-horizon temporal structure appropriate to a one-day forecast horizon and a seven-day input window $T = 7$, including autoregressive demand signals and gap-aware offsets that respect irregular calendar spacing. Second, it represents both node-local load state and national system context within each daily record, enabling the model to relate regional forecasts to grid-wide limitation and reserve conditions. Third, it enforces a strict leakage policy: same-day Operational Stress Index values are excluded from the model input specification even though an OSI-derived feature is computed for target construction and diagnostic use, consistent with Section III-A.

Raw variables versus engineered features. Raw variables comprise the division-level evening-peak demand, supply, and load measurements, national generation and demand scalars, operational limitation indicators, temperature, and calendar descriptors described in Section III-B. Preprocessing supplies encoded and standardised forms of these fields without creating new attributes. Engineered features are deterministic functions of raw and preprocessed quantities—lags, rolling statistics, normalised shares, grid aggregates, cyclical calendar encodings, and composite stress indicators—that are computed on the chronological timeline and merged with the baseline representation by date. Continuous engineered quantities are standardised using training-partition statistics only; a binary shedding-presence indicator is retained without scaling.

Temporal feature engineering. Four global temporal descriptors encode seasonality, long-run drift, and irregular sampling. Day-of-year sine and cosine features provide smooth annual periodicity without discontinuities at year boundaries. A monotonic trend index captures slow structural change across the multi-year record. An inter-observation gap feature records the elapsed days between consecutive observed records, explicitly signalling calendar discontinuities rather than treating missing days as zero-load intervals. At each node, three lag families supply autoregressive structure: one-day and seven-observation demand lags, and one-day load lags. Lags use observed-row offsets on the sorted timeline rather than fixed calendar offsets, so a seven-observation lag advances seven recorded days even when calendar gaps intervene. Seven-day demand rolling means are computed over the seven preceding observed rows with the current day excluded, yielding a past-only smoothed load level that complements point lags for short-horizon forecasting.

Node-level engineered features. The node input specification comprises three conceptual groups. A contemporaneous power-balance group carries processed demand, supply, and load at each division, anchoring the node to its current operational state. An autoregressive temporal group supplies demand lags at one and seven observations, a one-observation load lag, and a seven-observation demand rolling mean, encoding short-term persistence and weekly structure aligned with the input window. A regional derived group expresses each division's relative role in the national system: demand share $D_r(t)/\sum_j D_j(t)$, which normalises heterogeneous scales to a common simplex representation, and load intensity $L_r(t)/\max(D_r(t), \varepsilon)$ with $\varepsilon = 1$ MW, which expresses unmet demand relative to local load without treating sparse shedding zeros as informative magnitude on their own. These three groups yield $F_n = 9$ channels per region in the node tensor $\mathbf{X}^{\mathrm{node}}$.

Global engineered features. The global input specification comprises six conceptual groups. A temporal and calendar group includes the four engineered descriptors above—cyclical day-of-year encodings, trend index, and inter-observation gap—together with a holiday-category summary derived by consolidating the four one-hot holiday-category channels produced in Section III-C into a single ordinal indicator for model input. A grid-aggregate group sums total regional demand and total regional load and computes generation reserve as the difference between highest available generation and evening-peak demand at the generation endpoint, summarising system-wide balance and headroom. An operational group aggregates gas, coal, water, and maintenance limitation indicators into total operational limitation, records whether any division reports positive shedding, and—during feature construction only—forms a composite stress score from shedding intensity, reserve margin, and limitation pressure using train-split normalisation bounds; this composite is excluded from model inputs to prevent same-day OSI leakage when forecasting $\mathrm{OSI}(t+1)$. A weather group expresses temperature anomaly relative to train-estimated monthly means. A limitation group retains the four preprocessed operational-constraint scalars from the baseline representation. A national generation group retains two preprocessed generation-endpoint scalars. Together, these groups yield $F_g = 17$ channels in the global tensor $\mathbf{X}^{\mathrm{global}}$ at each timestep.

Operational feature representation. Operational engineering targets the constraint drivers underlying Task 2 without substituting the supervised OSI label into the input set. Total operational limitation condenses fuel, infrastructure, and maintenance constraints into a single MW-scale pressure index. Generation reserve links available capacity to contemporaneous demand. The binary any-division shedding indicator flags system-wide unmet-demand events that may precede graded stress escalation. The engineered stress composite mirrors the component structure of Section III-A—shedding ratio, reserve shortfall, and limitation pressure—but is withheld from $\mathbf{X}^{\mathrm{global}}$ at inference time so that stress predictions at $t + 1$ cannot access a normalised stress score already computed at

t.

Feature grouping. The implemented channels are organised into eleven coalitions that separate raw regional blocks from engineered groups: regional demand, supply, and load blocks (G1–G3); engineered lags and rolling statistics (G4); regional share and intensity features (G5); calendar and trend descriptors (G6); grid aggregates (G7); the limitation stack (G8); weather anomaly (G9); national generation scalars (G10); and the shedding indicator (G11). This grouping maps each coalition to either node or global scope while preserving the distinction between contemporaneous measurements and derived predictors. The coalition definitions support grouped attribution analysis in Section V (Table VII).

Leakage-aware feature design. All lag and rolling computations apply strictly backward shifts on the chronological timeline. Monthly temperature means, OSI normalisation bounds for the composite stress feature, and standardisation parameters for continuous engineered fields are estimated on the training partition only. The Operational Stress Index at day t is never included among the seventeen global model inputs. Demand and load histories through day t remain admissible for forecasting at $t+1$. Calendar gaps are propagated through the gap-day feature and gap-aware lags rather than imputed.

Final feature representation. At each forecast origin, the model receives a lookback window of $T = 7$ days. Node inputs form $\mathbf{X}^{\text{node}} \in \mathbb{R}^{T \times N \times F_n}$ with $N = 9$ and $F_n = 9$; global inputs form $\mathbf{X}^{\text{global}} \in \mathbb{R}^{T \times F_g}$ with $F_g = 17$, instantiating the input tensors defined in Section III-A.

E. Graph Construction

Feature tensors encode temporal signals but not spatial coupling among regional loads. Bangladesh evening-peak demand exhibits strong inter-division correlation from shared national growth, seasonality, and calendar shocks. We adopt a graph with each division as a vertex and weighted edges encoding historical demand co-movement, yielding a fixed spatial prior at every forecast origin.

Node definition. The vertex set $\mathcal{V}$ contains $N = 9$ nodes, one per administrative division—Barishal, Chattogram, Cumilla, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, and Sylhet—in fixed alphabetical order for reproducible indexing. Each node is aligned with the corresponding row of the node feature tensor $\mathbf{X}^{\text{node}}$ described in Section III-D: contemporaneous demand, supply, and load; autoregressive lags and rolling means; and regional share and intensity descriptors. National and system-level quantities remain global context in $\mathbf{X}^{\text{global}}$ rather than separate graph vertices, preserving a clear separation between node-local load state and grid-wide conditioning.

Spatial dependency modelling. Spatial dependence is modelled through undirected edges whose weights reflect pairwise statistical association in regional demand. During graph design, three construction strategies were specified—geographical adjacency, correlation-thresholded adjacency, and a hybrid of both—so that domain priors and data-driven coupling could be compared under a common node set. For the proposed framework, the correlation-only strategy is adopted as the definitive edge rule; alternative formulations are described in a later subsection, and the correlation-graph construction steps are detailed below.

Correlation computation. Pairwise edge weights are derived from Pearson correlation coefficients ρ_{ij} computed on regional evening-peak demand series using training-partition observations only ($n = 1{,}295$ daily records). Restricting correlation estimation to the training partition prevents held-out periods from influencing the spatial prior. Demand was selected as the correlation basis because it is the primary Task 1 target and exhibits the strongest and most stable inter-regional association in the exploratory record. Train-period demand correlations are uniformly positive across division pairs, with the weakest associations confined to a small subset of pairs; most divisions co-vary strongly over shared seasonal and growth cycles rather than evolving independently.

Edge definition and sparsification. An undirected edge (i, j) is retained when $\rho_{ij} \geq \tau$, with correlation threshold $\tau = 0.65$. Edge weight equals ρ_{ij} for retained pairs and zero otherwise. The threshold follows the correlation cut-off adopted during exploratory analysis and graph design: it retains pairs with substantively positive demand co-movement, excludes the weakest associations, and yields a connected graph suitable for spatial message passing. Under this rule, 33 of 36 possible undirected pairs are retained (91.7% pair density). Self-loops are excluded by setting diagonal entries to zero, so each node does not receive a trivial self-message through the spatial prior. The resulting topology is symmetric, non-negative, and connected.

Graph normalisation. Raw correlation weights are row-normalised to produce the adjacency matrix $\mathbf{A}$ used in the proposed framework. Each row is divided by its sum so that outgoing edge weights from a node form a convex combination summing to unity. From a methodological standpoint, row normalisation expresses each division's spatial neighbourhood as a weighted mixture of its neighbours rather than an unscaled sum of correlation magnitudes, which stabilises cross-division aggregation when node degree and edge strength vary. It also prevents high-degree nodes from dominating neighbour contributions solely because they retain more edges. The normalised matrix is fixed after training-partition estimation and applied unchanged across validation and test periods.

Static versus dynamic graph. The adjacency matrix is static: edge structure and weights are estimated once from the full training timeline and held fixed thereafter. Time-varying graph refresh—for example, through rolling correlation windows—was considered during graph design but deferred to keep the spatial prior reproducible and leakage-safe. Dynamic national context instead enters through the global feature channels and the temporal dimension of $\mathbf{X}^{\text{node}}$, while $\mathbf{A}$ supplies a stable map of historical inter-division demand coupling.

Alternative graph formulations. The geographical graph connects border-adjacent divisions with uniform binary edges (21 undirected pairs; 58.3% density), row-normalised for aggregation. This variant encodes an administrative-border prior on plausible power-exchange neighbourhoods but assigns equal weight to all geographic links regardless of observed coupling strength. The hybrid graph retains an edge when either a geographical link exists or $\rho_{ij} \geq 0.85$, with weight ρ_{ij} (24 undirected pairs; 66.7% density). This variant augments the geographic skeleton with the strongest non-border demand associations. Neither alternative enters the proposed framework, which employs the correlation-only rule with $\tau = 0.65$ described above.

Final graph representation. Graph construction yields row-normalised adjacency $\mathbf{A} \in \mathbb{R}^{N \times N}$ and a derived attention-bias matrix (log-scaled positive entries, masked absent edges) for additive spatial masking. Both matrices are fixed, train-estimated artefacts aligned to alphabetical node order and accompany each forecast window.

F. Proposed STGT Architecture

The Correlation-Aware Multi-Task Forecasting Framework—implemented as Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT)—integrates graph structure and feature tensors in one model that jointly forecasts next-day regional demand and OSI (Fig. 1). A parallel dual-branch design combines graph-aware spatial attention for inter-division coupling with a temporal transformer for within-division dynamics; learnable gating fuses the branches before task-specific heads.

Model inputs and representational layers. Three input layers feed the architecture, and each serves a distinct role. The data representation comprises the node tensor $\mathbf{X}^{\text{node}} \in \mathbb{R}^{T \times N \times F_n}$ and the global tensor $\mathbf{X}^{\text{global}} \in \mathbb{R}^{T \times F_g}$, constructed as described in Section III-D for a lookback window of $T = 7$ days ending at forecast origin t. The graph representation comprises the row-normalised adjacency matrix $\mathbf{A}$ and the derived attention-bias matrix from Section III-E; these artefacts are fixed across forward passes and do not vary with the input window. The neural architecture maps embedded inputs through spatial and temporal encoders, fuses the resulting hidden states, and decodes task outputs. Supervision targets—regional demand at $t+1$ and $\text{OSI}(t+1)$—are not model inputs; the leakage policy in Section III-A excludes same-day OSI from the feature set when forecasting next-day stress.

Input embedding. Node and global tensors are projected into a shared latent space d_{model}. Node channels receive a learnable regional identity embedding; global channels broadcast national calendar, limitation, and generation context at each timestep; positional encoding marks lookback position. The resulting $\mathbf{H}_0 \in \mathbb{R}^{T \times N \times d_{\text{model}}}$ feeds both encoder branches.

Graph integration. The correlation graph enters the spatial encoder through an additive attention bias derived from $\mathbf{A}$. For each ordered node pair (i, j), positive adjacency weights contribute a log-scaled bias term to the attention score, while absent edges receive a large negative mask so that message passing is restricted to the correlation-supported neighbourhood. This mechanism couples learned attention patterns with the train-estimated spatial prior from Section III-E rather than relying on free-form pairwise weights alone. Graph integration is applied independently at each of the T timesteps in the spatial branch, allowing inter-division information exchange to vary day by day within the window while the underlying adjacency structure remains static.

Spatial encoder. The spatial branch is a graph transformer [7], [17] that applies multi-head self-attention over the N nodes at each timestep. Given hidden states at timestep t, attention scores combine scaled query–key inner products with the graph bias matrix, and the softmax-normalised weights aggregate value vectors across nodes:

$$\text{Attn}_{\text{spatial}}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}} + \mathbf{B}_{\text{adj}}\right)\mathbf{V},$$

where $\mathbf{B}_{\text{adj}}$ denotes the attention-bias matrix derived from $\mathbf{A}$. Stacked encoder layers with residual connections and position-wise feed-forward transformations refine $\mathbf{H}_0$ into a spatially encoded sequence $\mathbf{H}_{\text{spatial}} \in \mathbb{R}^{T \times N \times d_{\text{model}}}$. A graph transformer was selected over fixed-weight graph convolutions because division coupling strength varies across pairs and because attention weights provide an interpretable record of spatial information flow. The spatial encoder addresses the limitation of treating regions as independent: it propagates demand-relevant signals along correlation-weighted edges at each day in the lookback window.

Temporal encoder. The temporal branch is a transformer encoder [17] applied independently to each node's T-step sequence, with parameters shared across nodes for efficiency. At each node, multi-head self-attention operates over timesteps, capturing short-range autoregressive structure and weekly periodicity aligned with the seven-day input window and lag-based features in Section III-D. The encoder outputs $\mathbf{H}_{\text{temporal}} \in \mathbb{R}^{T \times N \times d_{\text{model}}}$ with the same shape as the spatial branch. A full transformer encoder was preferred over a single shallow attention layer because the operational record exhibits multi-day seasonal structure within the window. Causal masking is not applied: the window contains only observations at or before the forecast origin, so ex-post encoding of the full lookback is consistent with the day-ahead forecasting setting.

Feature fusion. Spatial and temporal branches encode complementary aspects of the input—cross-division coupling and within-division dynamics—and are integrated through parallel fusion rather than a strict sequential pipeline. A learnable gate computed from the concatenated branch outputs forms a convex combination at each timestep and node:

$$\mathbf{H}_{\text{fused}} = \mathbf{G} \odot \mathbf{H}_{\text{spatial}} + (\mathbf{1} - \mathbf{G}) \odot \mathbf{H}_{\text{temporal}},$$

where $\mathbf{G} = \sigma(\text{Linear}([\mathbf{H}_{\text{spatial}}; \mathbf{H}_{\text{temporal}}]))$ and $\odot$ denotes element-wise multiplication. Parallel fusion allows the model to emphasise spatial or temporal evidence depending on context—for example, synchronised national load movement versus division-specific short-term persistence—without forcing one inductive bias to precede the other. Sequential spatial-then-temporal or temporal-then-spatial designs were rejected during architecture design because they delay one form of coupling and can blur the temporal structure before cross-division mixing, or defer spatial propagation of same-day regional shocks.

Shared latent representation. The fused sequence is reduced to a per-node shared embedding by selecting the last observed timestep in the window:

$$\mathbf{H}_{\text{shared}} = \mathbf{H}_{\text{fused}}[:, T, :, :] \in \mathbb{R}^{N \times d_{\text{model}}}.$$

This tensor constitutes the common latent state from which both tasks are decoded. A graph-level readout vector is additionally formed by mean-pooling $\mathbf{H}_{\text{shared}}$ across nodes and concatenating the result with the embedded global context at the final timestep, yielding a compact system-wide summary for stress prediction. The shared representation enforces multi-task information sharing: demand and OSI heads draw on the same spatio-temporal encoding of the input window rather than independent models that would duplicate feature extraction.

Demand prediction head. Task 1 is decoded through a dedicated per-node linear head that maps each row of $\mathbf{H}_{\text{shared}}$ to a scalar next-day demand forecast in megawatts, producing the regional output vector $\hat{\mathbf{y}}^{\text{demand}} \in \mathbb{R}^N$ defined in Section III-A. A node-specific linear mapping is sufficient because $\mathbf{H}_{\text{shared}}$ already encodes division identity through the regional embedding and spatial attention; the head's role is to project the shared latent state onto the MW scale for each region. This design yields nine heterogeneous forecasts from one trunk while preserving parameter efficiency relative to nine independent sequence models.

OSI prediction head. Task 2 is decoded through a graph-level head that consumes three complementary views of the shared state: the mean-pooled graph summary concatenated with the final-timestep global embedding, and the flattened node-level shared representation across all divisions. A feed-forward network with a bounded output activation maps this combined vector to a single scalar $\hat{y}^{\text{osi}} \in [0, 1]$. The OSI head requires both aggregated system context and division-resolved latent states because operational stress in this formulation is a national score influenced by spatially distributed load and constraint patterns. Restricting the stress decoder to graph-pooled information alone would discard node-level variation retained in $\mathbf{H}_{\text{shared}}$; restricting it to a single node would fail to represent system-wide stress.

Multi-task learning mechanism. Demand and OSI share embedding, both encoders, fusion, and $\mathbf{H}_{\text{shared}}$; only decoding layers differ. Shared encoding exploits common calendar, limitation, and load history while preserving distinct output semantics (node-resolved MW demand versus graph-resolved bounded OSI). Loss balancing is specified in Section III-H.

Architectural rationale. PF-STGT addresses three requirements: correlation-only graph conditioning matched to strong inter-regional demand coupling; parallel spatial and temporal transformer branches for cross-division and within-division dynamics; and a shared latent core with specialised demand and stress heads for joint day-ahead forecasting from one spatio-temporal pass.

G. Training Strategy

PF-STGT is trained on chronologically partitioned samples under the leakage policies of Sections III-A and III-C. Future observations never influence past inputs; preprocessing and graph parameters are frozen from training data; model selection uses validation only. Sliding windows with aligned multi-task targets convert the feature timeline and static graph into supervised examples without crossing partition boundaries.

Chronological partition. The operational timeline is divided into training, validation, and test subsets by ascending date, following the chronological split applied during preprocessing. The training partition spans 21 November 2019 to 15 June 2023 (1,295 daily records); validation spans 16 June 2023 to 19 March 2024 (277 records); and test spans 20 March 2024 to 30 December 2024 (278 records). Partition boundaries are strictly non-overlapping: the latest training date precedes the earliest validation date, and the latest validation date precedes the earliest test date. Chronological partitioning is required because random shuffling of days would place future grid states in the training set and inflate performance estimates. The split ratios and date ranges are summarised in Table I.

Sequence generation and warm-up exclusion. Training examples are generated independently within each partition. For a partition containing R chronologically ordered daily records, a valid forecast origin must satisfy two constraints. First, the lookback window of length $T = 7$ must lie entirely within the partition and contain only rows with complete engineered features. Second, the supervision index must lie one horizon step ($H = 1$) ahead of the window end and remain inside the same partition. The first seven rows of each partition are excluded as warm-up because seven-observation demand lags and seven-day rolling means are undefined at the series start; this policy aligns with the lag-based feature design in Section III-D. The resulting numbers of valid sliding-window samples per partition are reported in Table I. Calendar gaps in the source timeline are preserved: windows span consecutive observed rows within a partition rather than imputed calendar days.

Sliding-window construction. Each training example is built by sliding a fixed-length window one observed row at a time. For forecast origin index t_{end}, the input window covers days $[t_{\text{end}} - T + 1,\, t_{\text{end}}]$ and supplies the node tensor $\mathbf{X}^{\text{node}}$ and global tensor $\mathbf{X}^{\text{global}}$ described in Section III-D. The static adjacency matrix $\mathbf{A}$ and its derived attention-bias matrix from Section III-E are attached to every sample because the graph prior does not vary across origins. Window construction is identical across partitions; only the allowable range of t_{end} changes with partition size and boundary position. Sliding-window enumeration maximises the number of supervised examples available within each period without altering the one-day forecast horizon.

Multi-task target preparation. Supervision is prepared separately from model inputs. For Task 1, regional evening-peak demand at day $t_{\text{end}} + H$ is extracted from the cleaned interim timeline in raw megawatts for each of the nine divisions, producing the target vector $\mathbf{y}^{\text{demand}} \in \mathbb{R}^N$. For Task 2, the Operational Stress Index at the same future day is computed from the component formulae in Section III-A using train-partition min–max bounds for normalisation; the resulting scalar $\text{OSI}(t_{\text{end}} + H)$ lies in $[0, 1]$. Demand targets use unscaled MW values from the clean operational record so that the demand head learns on a physically interpretable scale. OSI component bounds are fit once on the training partition and frozen, consistent with the leakage policy that forbids using validation or test statistics in target construction. Same-day OSI at t_{end} is never included among input features when predicting $\text{OSI}(t_{\text{end}} + H)$.

Batch formation. Training iterates over mini-batches of windowed samples drawn from the training partition. Within a batch, node and global feature tensors are stacked along the batch dimension; demand and OSI targets are stacked correspondingly. The adjacency and attention-bias matrices are shared across all samples in a batch because the graph is static. Training batches are shuffled each epoch to vary the order of independent window samples; this shuffling does not alter the chronological ordering of observations within any individual window. Validation and test loaders preserve chronological sample order so that temporal structure remains inspectable. Batch construction therefore separates variable per-origin features from fixed graph artefacts shared across the national grid.

Validation protocol. During training, model parameters are updated using only the training partition. After each epoch, performance is assessed on the validation partition using the same windowing and target rules, without parameter updates. Validation metrics include macro demand mean absolute error over the nine regions and stress mean absolute error for the graph-level OSI target. For the multi-task PF-STGT, checkpoint selection is guided by a balanced composite validation score that reflects both demand and stress prediction error, so that model selection does not optimise a single task at the expense of the other. The test partition is not used during training, hyperparameter selection, or early stopping; it is reserved for the held-out evaluation reported in Section V.

Early stopping and checkpoint selection. Training stops when the composite validation score fails to improve for a fixed patience interval. The checkpoint with the lowest composite validation score is retained for test evaluation.

Reproducibility considerations. A fixed random seed governs initialisation and shuffling; frozen datasets, features, and graph specifications are verified before training; and partition boundaries, window length, horizon, and warm-up rules are applied identically across loaders. Table II summarises frozen hyperparameters and the multi-task loss specification.

H. Loss Function and Optimization

This subsection specifies the objective function and optimisation procedure. Minimisation uses a weighted multi-task loss over training mini-batches. Demand targets remain in megawatts at evaluation; the training objective applies scale-aware transformation so demand and stress gradients remain commensurate. Updates use Adam with weight decay, ReduceLROnPlateau scheduling, and gradient-norm clipping.

Multi-task objective. Let $\hat{\mathbf{y}}^{\text{demand}} \in \mathbb{R}^N$ denote the vector of predicted regional evening-peak demands and $\hat{y}^{\text{osi}} \in [0, 1]$ the predicted Operational Stress Index for a single forecast origin, with corresponding targets $\mathbf{y}^{\text{demand}}$ (MW) and y^{osi} prepared as in Section III-G. The joint training objective extends the formulation introduced in Section III-A by separating raw demand fit from stress fit and introducing an explicit demand-scale normalisation before task weighting:

$$\mathcal{L} = \frac{\mathcal{L}_{\text{demand}}^{\text{raw}}}{100} + \lambda_2 \, \mathcal{L}_{\text{osi}},$$

where $\lambda_2 = 20$ is the frozen stress-task weight and the divisor 100 (MW) maps the demand loss term to a magnitude comparable with the bounded OSI loss. Minimisation is performed over all trainable parameters $\boldsymbol{\theta}$ of the PF-STGT model by mini-batch optimisation using the Adam optimiser [28] with batch size $B = 32$.

Demand loss. Regional demand errors are aggregated with a macro-averaged Huber loss over the $N = 9$ divisions:

$$\mathcal{L}_{\text{demand}}^{\text{raw}} = \frac{1}{N} \sum_{r=1}^{N} \text{Huber}_{\delta}\left(\hat{y}_r^{\text{demand}},\, y_r^{\text{demand}}\right),$$

where $\text{Huber}_{\delta}(a, b)$ is the element-wise Huber function with threshold $\delta = 1.0$ MW:

$$\text{Huber}_{\delta}(a, b) = \begin{cases} \frac{1}{2}(a - b)^2, & |a - b| \le \delta, \\ \delta\left(|a - b| - \frac{\delta}{2}\right), & \text{otherwise.} \end{cases}$$

Huber regression was selected in preference to pure squared error because evening-peak demand exhibits occasional upper-tail excursions; the quadratic region penalises moderate errors smoothly, while the linear region limits the influence of large outliers that would otherwise dominate batch gradients. Macro-averaging across regions assigns equal weight to each division in the loss, consistent with the node-level forecasting objective in Section III-A. Targets remain in raw megawatts during training so that the demand head learns on the operational scale used for evaluation; only the aggregated loss contribution entering $\mathcal{L}$ is divided by 100 MW.

OSI loss. The stress head output passes through a sigmoid activation so that $\hat{y}^{\text{osi}}$ lies in $[0, 1]$, matching the bounded OSI target constructed in Section III-A. Stress fit is measured with mean squared error:

$$\mathcal{L}_{\text{osi}} = \frac{1}{B} \sum_{i=1}^{B} \left(\hat{y}_i^{\text{osi}} - y_i^{\text{osi}}\right)^2$$

at batch level (equivalently, MSE over scalar OSI predictions in each mini-batch). Squared error on a unit-interval target provides a smooth, well-conditioned regression signal for the graph-level stress scalar and aligns with the continuous OSI formulation adopted over binary or quantised stress labels.

Task weighting and scale balancing. Without rescaling, $\mathcal{L}_{\mathrm{demand}}^{\mathrm{raw}}$ typically attains values on the order of tens of megawatts, whereas $\mathcal{L}_{\mathrm{osi}}$ remains close to zero because OSI targets are normalised to $[0, 1]$. A direct sum of the two terms would therefore allocate almost all gradient magnitude to the demand head, leaving the stress head under-trained despite joint architectural sharing. Dividing $\mathcal{L}_{\mathrm{demand}}^{\mathrm{raw}}$ by 100 MW compresses the demand term to a $\mathcal{O}(1)$ scale appropriate for combination with MSE on OSI. The weight $\lambda_2 = 20$ further elevates the stress contribution so that both task heads receive comparable optimisation pressure while preserving megawatt-scale demand targets. The composite validation score used for early stopping in Section III-G applies the same 100 MW scaling to demand error, keeping model selection consistent with the training objective. The chosen λ_2 and normalisation divisor are fixed prior to training and are not tuned on the test partition.

Optimizer and weight decay. Parameters are updated with the Adam optimiser [28] at learning rate $\eta = 5 \times 10^{-4}$ and weight decay $\lambda_{\mathrm{wd}} = 10^{-4}$. Adam provides per-parameter adaptive step sizes that accommodate the heterogeneous gradient magnitudes arising from the parallel spatial and temporal transformer branches and the two task-specific heads. Weight decay supplies mild ℓ_2 regularisation on trainable weights, discouraging overfitting on the finite daily training window without altering the loss formulation explicitly. Optimisation runs for up to 200 epochs, subject to the early-stopping rule in Section III-G.

Learning-rate schedule. After each training epoch, the learning rate is adjusted by a ReduceLROnPlateau scheduler that monitors validation macro demand MAE. When this metric fails to improve for five consecutive epochs, the learning rate is multiplied by a factor of 0.5. Validation demand MAE is preferred for learning-rate adaptation because it varies smoothly across epochs and supplies a stable optimisation signal on the chronologically ordered validation set; model selection and early stopping nonetheless rely on the composite validation score defined in Section III-G, so checkpoint retention reflects balanced multi-task performance rather than demand fit alone. The scheduler therefore operates independently of the early-stopping criterion, allowing learning-rate decay to proceed even when the composite score has not yet triggered termination.

Gradient clipping. Global gradient-norm clipping with maximum norm 1.0 is applied immediately before each optimiser step. Clipping bounds the ℓ_2 norm of the concatenated parameter gradient vector, mitigating occasional large back-propagation steps that can arise from outlier demand residuals or from attention layers in the graph transformer branch. This stabilises training across epochs without changing the defined loss function.

TABLE I
DATASET SUMMARY.

Property	Value
Domain	Bangladesh national power grid (9 divisions)
Input window (T)	7 days
Forecast horizon	1 day
Node features	9
Global features	17
Train windows	1,281
Validation windows	263
Test windows	264

TABLE II
FROZEN TRAINING CONFIGURATION (S2 / A6).

Setting	Value
Loss	$\mathcal{L} = \mathrm{Huber(demand)}/100 + \lambda_2 \cdot \mathrm{MSE(OSI)}$
λ_2	20.0
Optimiser	Adam, lr $= 5 \times 10^{-4}$, wd $= 10^{-4}$
Batch size	32
Early stopping	Patience 15 (composite val. score)
Parameters	749,058

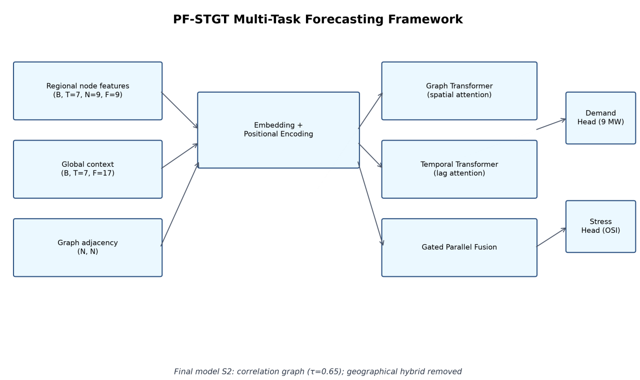

Fig. 1. End-to-end PF-STGT multi-task forecasting framework.

Numerical precision. Full-precision arithmetic is used throughout training. This methodological choice prioritises numerical stability on the modest daily sample size over the throughput gains of reduced-precision optimisation, which was not required for reliable convergence of the PF-STGT model.

Optimisation rationale. Huber macro-averaged demand loss limits outlier sensitivity; demand-loss normalisation and λ_2 prevent collapse to demand-only learning; and Adam, weight decay, plateau scheduling, gradient clipping, and full-precision arithmetic support stable convergence on fewer than 1,300 training windows. Frozen hyperparameters appear in Table II.

IV. EXPERIMENTAL SETUP

A. Experimental Environment

All experiments run in a version-controlled environment supporting reproducible training, evaluation, and post-hoc analysis of PF-STGT. The stack combines pinned Python dependencies with PyTorch deep learning, classical machine-

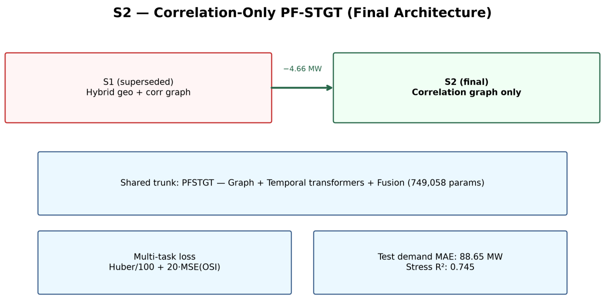

Fig. 2. Architecture freeze: S2 correlation-only graph PF-STGT.

learning routines, and explainability tooling. Hardware selection follows a portable policy (CUDA, Apple MPS, then CPU) so model definitions and data pipelines remain unchanged across platforms.

Compute platform. Deep-learning jobs select the execution device at runtime. Frozen PF-STGT runs—training, ablation, and explainability—used the **MPS** backend. Classical benchmarks (B01–B03) trained and evaluated on **CPU** without loading PyTorch.

Software environment. Experiments used Conda environment `smartgrid-stgt` (Python 3.11) with dependencies pinned in `requirements.txt` for the publication-freeze repository (25 June 2026). Core packages: **PyTorch 2.3.1** [29], **PyTorch Geometric 2.5.3** [30], **NumPy 1.26.4**, **Pandas 2.2.2**, **scikit-learn 1.5.0** [31], **SHAP 0.45.1** [18], and **Captum 0.7.0** [32]. Hydra and PyYAML handled configuration and logging.

Reproducibility controls. Versions are frozen in `requirements.txt` and `environment.yml`. Seed 42 governs initialisation, shuffling, and stochastic operations. Datasets, features, and graph specifications are verified against frozen inputs. Reported results derive from the publication-freeze repository and were not regenerated during manuscript preparation.

B. Baseline Models

We compare PF-STGT against seven reference models (B01–B07) [2], [3], [5], [21], [33] spanning classical machine learning, recurrent deep learning, and graph-based spatio-temporal learning. B01–B06 target Task 1 (regional demand); B07 supports joint Task 1 and Task 2 (OSI) as the historical PF-STGT reference. The final model S2 is evaluated separately in the ablation study.

Classical baselines (B01–B03). *B01 (Linear Regression)* flattens the seven-day feature window and applies ordinary least-squares multi-output regression. *B02 (Random Forest)* [2] uses bagged decision trees for non-linear interactions. *B03 (XGBoost)* [3] represents gradient boosting. Each classical model also fits a scalar OSI regressor for common stress evaluation.

Deep temporal baselines (B04–B05). *B04 (LSTM)* [5] and *B05 (GRU)* [33] use two-layer recurrent encoders (hidden 128, dropout 0.1) with concatenated node and global features and a linear head outputting nine regional forecasts.

Graph-based baseline (B06). *B06 (T-GCN)* [21] applies single-hop graph convolution with the row-normalised hybrid adjacency matrix and GRU temporal aggregation.

Multi-task PF-STGT reference (B07). *B07 (PF-STGT W20)* implements parallel-fusion graph transformer training under the W20 protocol with hybrid geographical–correlation graph conditioning. S2 (correlation-only graph, ablation A6) supersedes B07 as the publication-facing architecture.

Fairness protocol. All models share chronological train/validation/test partitions (Section III, Table I), the frozen feature store, leakage policies, and windowing ($T = 7$, $H = 1$, seven-row warm-up). Graph models receive hybrid adjacency (B06, B07) or correlation adjacency (S2). Task 1 uses macro-averaged MAE, RMSE, MAPE, and R^2; Task 2 uses MAE, RMSE, and R^2 on OSI. Hyperparameters were tuned on validation only; seed 42 governs frozen runs.

C. Evaluation Metrics

Performance is assessed on the held-out test partition after model selection on training and validation data. Task 1 (regional evening-peak demand) is the primary axis; Task 2 (OSI) is reported for multi-task models.

Evaluation scope. Metrics are computed on 264 valid test windows (Table I) without refitting scalers, graph parameters, or weights.

Task 1 — Regional demand metrics. For region $r \in \{1, \ldots, 9\}$, let $D_{r,i}$ and $\hat{D}_{r,i}$ denote observed and predicted evening-peak demand (MW) on test sample i.

$$\mathrm{MAE}_r = \frac{1}{n}\sum_{i=1}^{n}\left|D_{r,i} - \hat{D}_{r,i}\right|, \qquad \mathrm{MAE}_{\mathrm{macro}} = \frac{1}{N}\sum_{r=1}^{N}\mathrm{MAE}_r, \tag{3}$$

$$\mathrm{RMSE}_r = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(D_{r,i} - \hat{D}_{r,i}\right)^2}, \quad \mathrm{RMSE}_{\mathrm{macro}} = \frac{1}{N}\sum_{r=1}^{N}\mathrm{RMSE}_r. \tag{4}$$

Macro demand MAE is the **primary ranking metric**. RMSE accompanies MAE to highlight occasional large misses.

Mean absolute percentage error (MAPE) is

$$\mathrm{MAPE} = \frac{100}{|\mathcal{V}|}\sum_{(r,i)\in\mathcal{V}}\frac{\left|D_{r,i} - \hat{D}_{r,i}\right|}{|D_{r,i}|},$$

where $\mathcal{V}$ contains valid region-day pairs with $|D_{r,i}| > \varepsilon$. MAPE supports literature comparison but is not used for selection.

Coefficient of determination (R^2) is computed per region and macro-averaged:

$$R_r^2 = 1 - \frac{\sum_{i=1}^{n}\left(D_{r,i} - \hat{D}_{r,i}\right)^2}{\sum_{i=1}^{n}\left(D_{r,i} - \bar{D}_r\right)^2}, \qquad R^2_{\mathrm{macro}} = \frac{1}{N}\sum_{r=1}^{N}R_r^2.$$

Macro R^2 weights divisions equally. Dhaka MAE is reported separately given its share of national evening-peak load.

Task 2 — Operational stress metrics. OSI metrics apply over n test samples:

$$\mathrm{MAE}_{\mathrm{osi}} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i^{\mathrm{osi}} - \hat{y}_i^{\mathrm{osi}}\right|, \quad \mathrm{RMSE}_{\mathrm{osi}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i^{\mathrm{osi}} - \hat{y}_i\right.}\,,\;,$$

$$R^2_{\mathrm{osi}} = 1 - \frac{\sum_{i=1}^{n}\left(y_i^{\mathrm{osi}} - \hat{y}_i^{\mathrm{osi}}\right)^2}{\sum_{i=1}^{n}\left(y_i^{\mathrm{osi}} - \bar{y}^{\mathrm{osi}}\right)^2}.$$

Pearson correlation on OSI is supplementary only. Inferential comparison uses paired Wilcoxon signed-rank tests [34] with Bonferroni correction [35] on per-window macro demand MAE (Section IV-E).

D. Hyperparameter Configuration

Hyperparameters were selected on validation only; the test split was not used for tuning. Seed 42 governs stochastic components.

Shared windowing. All models use $T = 7$ days and $H = 1$.

Proposed PF-STGT (S2 / A6). Table II lists optimisation settings: Adam [28] ($\eta = 5\times10^{-4}$, weight decay 10^{-4}), batch size 32, up to 200 epochs with early stopping (patience 15 on balanced validation score), ReduceLROnPlateau on validation macro demand MAE (factor 0.5, patience 5), gradient clipping at 1.0, $\lambda_2 = 20.0$, and demand-loss scaling dividing Huber loss by 100 MW. Capacity: $d_{\mathrm{model}} = 128$, four heads, two spatial and two temporal layers, feed-forward dimension 256, dropout 0.1.

Deep baselines (B04–B06). B04/B05 use two-layer LSTM/GRU (hidden 128, dropout 0.1); B06 uses hidden 64. Trained with AdamW [36], batch size 32, up to 60 epochs, early stopping on validation demand MAE.

Classical baselines (B01–B03). B01 uses ordinary least squares; B02 uses 300 trees; B03 uses 300 estimators (max depth 6, learning rate 0.1). Inputs are standardised from training-partition statistics only.

E. Statistical Testing

Inferential analysis targets Task 1 macro demand MAE across three families: benchmark (B07 versus B01–B06), ablation (A1 versus A2–A6), and architecture simplification (S1 versus S2–S4).

Paired comparison unit. For test window i, per-window macro MAE is $\mathrm{MAE}_i = \frac{1}{N}\sum_{r=1}^{N}|D_{r,i} - \hat{D}_{r,i}|$. Paired differences $\Delta\mathrm{MAE}_i$ control for day-specific conditions.

Wilcoxon signed-rank test. The paired Wilcoxon test [34] is applied to $\{\Delta\mathrm{MAE}_i\}$ with two-sided p-values by default; one-sided tests require a pre-registered directional hypothesis.

Multiple-comparison correction. Bonferroni thresholds: $\alpha_B = 0.05/6 \approx 0.0083$ for benchmarks; $\alpha_B = 0.01$ for ablation and architecture families.

Effect size and uncertainty. Cohen's d [37] and bootstrap 95% confidence intervals [38] (2,000 resamples, seed 42) supplement p-values. Significance indicates systematic per-window MAE difference; practical importance follows megawatt magnitude, bootstrap intervals, and effect-size conventions.

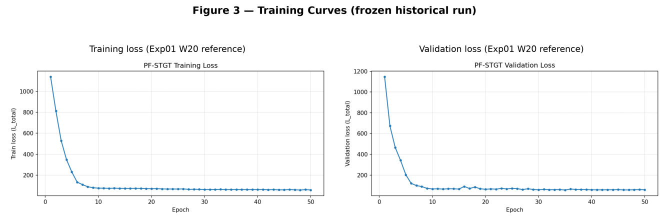

Fig. 3. Training and validation loss curves (W20 protocol reference).

V. Results

A. Overall Benchmark Performance

1) Overall benchmark summary: The benchmark evaluation compares S2 against seven reference models (B01–B07) on 264 held-out test windows (Table III). S2 attains the lowest macro demand MAE (88.65 MW), followed by B07 (93.31 MW) and B02 (97.03 MW). Recurrent and GCN baselines exceed 230 MW MAE. S2 also leads on OSI (MAE 0.0371, R^2 0.745), improving over B07 (0.0499 MAE, 0.585 R^2) and all stress-capable classical and deep baselines.

2) Demand forecasting performance: S2 achieves the lowest macro demand MAE (88.65 MW; RMSE 127.29 MW; MAPE 6.55%), improving 4.66 MW (5.0%) over B07 (93.31 MW) and 8.38 MW over B02 (97.03 MW). B03 records 109.73 MW. Recurrent baselines B04/B05 exceed 233 MW MAE with negative macro R^2; B06 (T-GCN) [21] reaches 257.21 MW. Graph-aware PF-STGT variants consistently outperform classical ensembles and standalone recurrent encoders on megawatt MAE.

B02 reports higher macro R^2 (0.984) than S2 (0.684) and B07 (0.674), but MAE remains the primary ranking metric—R^2 and MAE quantify different performance aspects. S2 represents the stronger demand forecaster among multi-task models in Table III.

3) Operational Stress prediction: S2 achieves stress MAE 0.0371 and R^2 0.745—the best OSI performance in Table III. B07 records 0.0499 MAE and 0.585 R^2; B02/B03 reach 0.0481/0.555 and 0.0497/0.525 respectively. B04–B06 yield negative stress R^2 (-0.191 to -0.304). S2 reduces stress MAE by 0.0128 and raises R^2 by 0.160 absolute versus B07, delivering the most informative day-ahead stress signal in the benchmark suite.

4) Statistical evidence: Paired Wilcoxon tests [34] on per-window macro demand MAE (n = 264) support benchmark rankings (Table IV). With B07 as reference and Bonferroni threshold [35] α = 0.0083, B07 significantly outperforms B01–B06; median paired improvements range from 4.92 MW (B02) to 160.66 MW (B06). S2 versus B07: median ΔMAE = -5.43 MW, p = 5.5×10^{-5}, bootstrap 95% CI [38] [-7.17, -2.16] MW. B07 versus B02 is significant (p = 0.00135, Cohen's d

[37] = -0.077) but the bootstrap CI [-8.87, 2.62] MW spans zero. No formal S2-versus-B02 test was conducted; the 8.38 MW point-estimate advantage should be interpreted alongside the B07–B02 result.

5) Overall interpretation: S2 leads Table III on both tasks: lowest macro demand MAE (88.65 MW) and highest stress R^2 (0.745). PF-STGT variants outperform standalone LSTM, GRU, and T-GCN architectures (demand MAE $>$230 MW; uninformative stress predictions). Classical ensembles remain competitive but do not match S2 megawatt accuracy despite B02's higher macro R^2. Demand gains over random forest are modest and should be communicated with appropriate statistical qualification.

B. Regional Forecast Analysis

1) Macro context and regional scope: Building on Section V-A, this subsection examines regional error distribution (Fig. 4). S2 per-region MAE spans 32.73 MW (Barishal) to 293.98 MW (Dhaka); macro summaries understate planning significance for the principal load centre.

2) Regional demand performance: Per-region MAE for S2 is tabulated in supplementary ablation outputs. The three lowest-MAE divisions—Barishal, Sylhet, and Rangpur—each remain below 55 MW, whereas Dhaka alone exceeds 290 MW and Khulna exceeds 88 MW. At the macro level, S2 records demand RMSE of 127.29 MW, demand MAPE of 6.55%, and macro demand R^2 of 0.684 on the test split.

Relative to B07 (Table S1), peripheral accuracy is broadly comparable but not uniform: S2 records slightly higher MAE in Barishal, Chattogram, Cumilla, Mymensingh, and Sylhet, while reducing MAE in Khulna (88.62 versus 118.72 MW), Rajshahi (76.25 versus 85.70 MW), Rangpur (53.97 versus 59.77 MW), and Dhaka (293.98 versus 299.78 MW). Against random forest B02, S2 achieves lower Dhaka and Khulna MAE while trailing B02 in Barishal and several other divisions by small margins. S2's macro leadership reflects division-level gains and minor regressions rather than uniform improvement. Planners should treat Fig. 4 as an equal-weight summary and consult Table S1 for load centres with the largest residuals.

3) Regional consistency and variability: Per-region MAE under S2 exhibits substantial cross-division spread: the ratio between Dhaka MAE (293.98 MW) and Barishal MAE (32.73 MW) exceeds ninefold on the test split. B07 and B02 display a similar pattern in Table S1, with Dhaka MAE of 299.78 MW and 311.59 MW respectively against peripheral MAE values generally below 90 MW. This consistency of error concentration across model families indicates that regional heterogeneity in absolute megawatt error is a persistent feature of the test period rather than an artefact of a single architecture.

Comparing S2 with B07 division by division, the largest improvements occur in Khulna (-30.10 MW), Rajshahi (-9.45 MW), Dhaka (-5.80 MW), and Rangpur (-5.80 MW), whereas S2 records slightly higher MAE than B07 in five peripheral divisions, each by less than 5 MW in absolute terms. The macro MAE gain of 4.66 MW (S2 versus B07) aggregates offsetting regional effects; deployment reviews should examine division-level metrics alongside Fig. 4.

Per-region R^2 values in Table S1 further illustrate metric divergence. For B07, division-level R^2 ranges from 0.4808 (Barishal) to 0.8270 (Rajshahi); for B02, Rajshahi reaches 0.8274 while Dhaka records 0.6666 despite B02's higher Dhaka MAE than S2. R^2 and MAE therefore need not agree at the division level, consistent with the macro-level distinction noted in Section V-A2. Forecast quality assessments that rely on a single metric may reach different conclusions depending on whether absolute megawatt deviation or variance explanation is emphasised.

4) Dhaka-specific performance: Dhaka carries the largest evening-peak load among the nine divisions and accounts for approximately 35.7% of mean national regional demand in the exploratory data analysis. On the test split, Dhaka also exhibits the highest demand variability among audited regions, with B07 recording an observed demand standard deviation of 670.15 MW in Table S1—far exceeding peripheral divisions. Dhaka MAE under S2 is 293.98 MW, compared with 299.78 MW for B07 and 311.59 MW for B02 (Table S1); the supplementary audit reports Dhaka R^2 of 0.5744 for B07 and 0.6666 for B02 on the same split.

S2 therefore achieves the lowest audited Dhaka MAE among the three models with published division-level tables, reducing absolute Dhaka error by 5.80 MW relative to B07 and 17.61 MW relative to B02. Because Dhaka's load scale dominates national totals, even modest percentage improvements translate into larger absolute megawatt reductions than comparable percentage gains in low-load divisions. Nevertheless, Dhaka MAE remains an order of magnitude above peripheral MAE under S2, so the principal national load centre continues to account for the largest share of forecast error in megawatt terms. Day-ahead planning for the capital region should continue to treat Dhaka as a separately reported diagnostic rather than relying solely on macro-averaged scores.

5) Regional strengths and limitations: The strongest regional performance under S2 occurs in Barishal, Sylhet, and Rangpur, where MAE remains below 55 MW, and in Mymensingh (59.22 MW). These divisions contribute relatively little to the macro numerator despite receiving equal weight in the macro average. In the present benchmark, S2 also delivers its largest MAE reductions relative to B07 in Khulna and Rajshahi, indicating that the proposed model improves selected mid-scale divisions with material evening-peak load without registering worse MAE in every peripheral node.

The principal regional limitation is persistent Dhaka error concentration (293.98 MW MAE). Secondary limitations include small MAE regressions versus B07 in five divisions (each below 5 MW) and absent per-region R^2 for S2. Overall, regional evidence supports S2 as a consistent multi-division forecaster with meaningful Dhaka gains, while equal-weight macro summaries in Fig. 4 understate the planning significance of the largest load centre.

C. Operational Stress (OSI) Analysis

1) Scope and evaluation context: Task 2 is evaluated on the same 264-window test split as demand forecasting. OSI is reported as MAE and R^2 on $[0, 1]$. Demand-only variant A4 (Table V) is excluded. Analysis combines external benchmark comparison (Table III) with configuration comparison (Table V; Fig. 5).

2) OSI benchmark comparison: S2 leads stress-capable models in Table III (OSI MAE 0.0371, R^2 0.745), improving over B07 (0.0499, 0.585) and B02/B03 (0.0481/0.555 and 0.0497/0.525). B04–B06 yield OSI MAE 0.0861–0.0891 with negative R^2 (-0.191 to -0.304); B01 records 0.1074 MAE and R^2 of -1.824. Relative to B07, S2 reduces OSI MAE by 0.0128 (25.7%) and raises R^2 by 0.160; relative to B02, by 0.0110 MAE and 0.190 R^2. Classical baselines remain closer on absolute OSI MAE than on variance explanation.

3) Configuration-level OSI performance: Table V reports OSI metrics for the frozen multi-task configuration suite evaluated on the identical test split. S2 (variant A6) records OSI MAE of 0.0371 and OSI R^2 of 0.745. The historical hybrid-graph reference A1 (equivalent to B07) records OSI MAE of 0.0499 and OSI R^2 of 0.585. Variants A2 (no graph) and A3 (no transformer) each achieve OSI MAE of 0.0405 and OSI R^2 of 0.701—lower absolute error and higher R^2 than A1 but still behind A6 on both stress metrics. Variant A5 (geographical graph only) attains the lowest OSI MAE among multi-task configurations at 0.0340 and the highest stress R^2 at 0.764, while recording demand MAE of 97.98 MW. Variant A4 (single-task) achieves the lowest demand MAE in the suite at 86.89 MW but provides no OSI output.

Fig. 5 orders the same configurations by test demand MAE: A4 at 86.89 MW, A6 (S2) at 88.65 MW, A3 at 92.64 MW, A1 at 93.31 MW, A2 at 93.93 MW, and A5 at 97.98 MW. Read together with Table V, the figure and table show that lower demand MAE does not imply superior OSI accuracy across configurations. A5 leads on both OSI metrics yet records the highest demand MAE among the multi-task variants shown in Fig. 5, whereas A6 combines the second-lowest demand bar in that chart with the second-best OSI MAE and R^2 in Table V. The benchmark artefacts demonstrate this demand–stress divergence but do not establish the underlying mechanism responsible for variant-level trade-offs. For joint day-ahead planning, S2 represents the configuration that simultaneously delivers the lowest multi-task demand MAE among OSI-producing variants in Fig. 5 and the strongest OSI R^2 in the external benchmark of Table III, accepting that A5 attains marginally lower OSI MAE and higher stress R^2 at the cost of 9.33 MW higher demand error relative to S2.

4) Statistical evidence and reporting limits: Formal inferential testing in the frozen statistical record is defined on per-window macro **demand** MAE rather than on OSI residuals. The paired Wilcoxon comparison of A1 versus A6 (S2) reports median demand ΔMAE of -5.43 MW, two-sided p = 5.5×10^{-5}, and bootstrap 95% confidence interval [-7.17, -2.16] MW on the test split (Table V; statistical summary). That interval excludes zero and corroborates the demand improvement of S2 over the historical reference; no corresponding Wilcoxon test, p-value, or bootstrap confidence interval for OSI MAE or OSI R^2 was computed in Experiments 02 or 03.

Stress improvement from A1 to A6 rests on point estimates only (OSI MAE 0.0499 to 0.0371; R^2 0.585 to 0.745); no paired OSI hypothesis tests were computed. Wilcoxon results in Table IV cover demand MAE only. OSI claims should be communicated as point-estimate advantages on the frozen test split.

5) Practical implications and limitations: S2 delivers graph-level stress forecasts with MAE below 0.04 and R^2 of 0.745, explaining approximately three-quarters of test-set OSI variance while leading the demand benchmark (Section V-A). For joint day-ahead load and stress monitoring, S2 offers the strongest credentials in Tables III and V, outperforming classical baselines on stress R^2 (B02: 0.555; B03: 0.525) and recurrent architectures with uninformative stress predictions.

Three limitations follow. First, no stress-specific confidence intervals or corrected hypothesis tests are available. Second, A5 achieves lower OSI MAE (0.0340) and higher stress R^2 (0.764) than S2 but at materially higher demand MAE (97.98 versus 88.65 MW). Third, A4 establishes a demand-only bound of 86.89 MW—1.76 MW below S2—without OSI output, confirming a demand–stress Pareto trade-off. Deployment should monitor reported MAE and R^2 alongside division-level demand metrics (Section V-B).

D. Ablation Study

1) Ablation scope and reference configuration: The frozen ablation programme evaluates six PF-STGT configurations (A1–A6) on the held-out test split of 264 forecast windows, with A1 serving as the reference: hybrid geographical–correlation graph, full transformer trunk, and multi-task demand–OSI learning under the W20 protocol. Table V consolidates test-set macro demand MAE, demand R^2, stress MAE, and stress R^2 for each variant; Fig. 5 visualises the same configurations ranked by demand MAE. Together, the table and figure permit component-wise comparison against A1 without rerunning experiments.

A1 (equivalent to B07) records demand MAE 93.31 MW, demand R^2 0.674, stress MAE 0.0499, and stress R^2 0.585. Subsequent contrasts use A1 as the frozen hybrid-graph baseline from which S2 was derived.

2) Graph contribution: Graph topology is isolated by three variants: A2 removes graph conditioning entirely while retaining the hybrid multi-task trunk; A5 replaces the hybrid adjacency with a geographical graph only; and A6 (S2) replaces it with a correlation-only graph (τ = 0.65). On demand MAE, A2 records 93.93 MW—0.62 MW above A1—whereas A5 records 97.98 MW, the highest demand error among the six configurations. A6 achieves 88.65 MW, reducing demand MAE by 4.66 MW relative to A1 (approximately 5.0%). Paired Wilcoxon testing reports median demand ΔMAE of +2.81 MW for A1 versus A2 (p = 0.301, bootstrap 95% CI [-3.69, 4.93] MW), +3.85 MW for A1 versus A5 (p = 1.48×10^{-4}, Bonferroni significant at α = 0.01, CI [2.19, 6.90] MW), and -5.43 MW for A1 versus A6 (p = 5.5×10^{-5}, CI [-7.17, -2.16] MW).

On the stress task, A2 improves upon A1 (stress MAE 0.0405 versus 0.0499; stress R^2 0.701 versus 0.585), whereas A5 attains the lowest stress MAE in Table V at 0.0340 with stress R^2 of 0.764—marginally above A6 (0.0371 MAE, 0.745 R^2). Graph choice affects both tasks non-uniformly: A2 shows no significant demand change versus A1; A5 significantly degrades demand; A6 significantly improves demand and raises stress R^2 versus A1. Correlation-graph conditioning (A6) is the only variant with statistically supported demand improvement over A1.

3) Transformer contribution: Variant A3 removes the transformer encoder while preserving the hybrid graph and multi-task heads. A3 records demand MAE of 92.64 MW—0.67 MW below A1—and stress MAE of 0.0405 with stress R^2 of 0.701, compared with A1 values of 93.31 MW, 0.0499, and 0.585 respectively. Wilcoxon comparison of A1 versus A3 yields median demand ΔMAE of -1.13 MW, two-sided p = 0.384, and bootstrap 95% CI [-2.35, 1.08] MW, which spans zero; the comparison is not significant at the Bonferroni-adjusted threshold α = 0.01.

On the hybrid graph, A3 matches A1 on demand MAE (non-significant Wilcoxon); transformer removal is not justified on hybrid-graph grounds without topology-specific validation.

4) Multi-task contribution: Multi-task learning is isolated by variant A4, which disables the OSI head and trains demand only on the hybrid graph. A4 achieves the lowest demand MAE in the ablation suite at 86.89 MW with demand R^2 of 0.731, compared with A1 at 93.31 MW and 0.674. Wilcoxon comparison of A1 versus A4 reports median demand ΔMAE of -5.25 MW, p = 0.00284, and bootstrap 95% CI [-10.63, -2.40] MW; A4 improves demand relative to the multi-task reference A1, though the Bonferroni flag for "significantly worse" does not apply because A4 is better, not worse. A4 provides no stress MAE or stress R^2 entries in Table V.

Relative to the final multi-task configuration A6 (S2), A4 retains a 1.76 MW demand advantage (86.89 versus 88.65 MW) but forfeits OSI forecasting entirely. Conversely, A6 improves demand by 4.66 MW over A1 while delivering stress R^2 of 0.745—0.160 above A1—indicating that multi-task training under the correlation graph does not reproduce the demand-only ceiling but does enable joint stress prediction at substantially higher quality than A1. Multi-task learning trades 1.76 MW demand MAE versus the single-task ceiling (A4) for OSI outputs with R^2 above 0.74; demand-only use cases may prefer A4, joint planning A6.

5) Demand–stress trade-offs across A1–A6: Fig. 5 orders all six configurations by demand MAE: A4 (86.89 MW), A6 (88.65 MW), A3 (92.64 MW), A1 (93.31 MW), A2 (93.93 MW), and A5 (97.98 MW). Read with Table V, the ranking shows that the lowest demand bar (A4) corresponds to the absence of stress outputs, while the second-lowest (A6) pairs the best multi-task demand score with stress R^2 of 0.745. A5 occupies the highest demand position yet attains the best stress MAE (0.0340) and stress R^2 (0.764) among OSI-producing variants, confirming that stress-optimal and demand-optimal configurations diverge within the frozen suite. A2 and A3 cluster near A1 on demand while recording identical stress MAE (0.0405) and stress R^2 (0.701), suggesting similar joint-task behaviour when either the graph module or transformer is removed on the hybrid trunk.

No single configuration minimises demand MAE and maximises both stress metrics; selection requires explicit joint versus demand-only priorities.

6) Final model selection: Architecture A6 (correlation-only graph, full transformer, multi-task) is designated S2—the final proposed model—based on its position in Table V and Fig. 5. A6 delivers the lowest demand MAE among all multi-task configurations (88.65 MW), with statistically supported improvement over A1 (median ΔMAE -5.43 MW, p = 5.5×10^{-5}, bootstrap 95% CI [-7.17, -2.16] MW). It simultaneously records stress R^2 of 0.745 and stress MAE of 0.0371, exceeding A1 on both stress metrics and matching the S2 row in the external benchmark of Table III. A6 is not demand-optimal within the ablation suite—A4 remains 1.76 MW lower—and is not stress-optimal among multi-task variants—A5 records lower stress MAE and higher stress R^2 at the cost of 9.33 MW additional demand error.

A6 (S2) is selected for best multi-task demand accuracy with strong OSI performance and significant improvement over A1, adopting correlation-only graph conditioning with full transformer capacity and W20 multi-task heads.

E. Error Analysis

1) Scope and aggregate error magnitudes: Sections V-A–V-D established rankings; this subsection characterises remaining error structure on the test split. The frozen evaluation record for S2 (variant A6) reports macro demand MAE of 88.65 MW, demand RMSE of 127.29 MW, demand MAPE of 6.55%, and macro demand R^2 of 0.684, together with OSI MAE of 0.0371, OSI RMSE of 0.0473, and OSI R^2 of 0.745 on 264 test windows (Table III; `ablation_raw.json`).

The ratio of demand RMSE to demand MAE is approximately 1.44 (127.29/88.65), indicating that squared-error emphasis falls on a subset of windows or regions with larger absolute deviations than the macro mean. For OSI, the RMSE-to-MAE ratio is approximately 1.27 (0.0473/0.0371), suggesting comparatively less tail amplification relative to demand on the normalised stress scale. Pooled signed-residual statistics—mean bias, skewness, and histogram shape—for S2 are not tabulated in the frozen results package; the planned error-analysis framework defined post-hoc residual protocols, but residual-store generation was not executed for the final checkpoint. Aggregate error magnitudes therefore rest on MAE, RMSE, and R^2 summaries rather than on full distributional diagnostics for S2.

2) Regional error concentration: Per-region demand MAE for S2 on the test split ranges from 32.73 MW (Barishal) to 293.98 MW (Dhaka), with Khulna at 88.62 MW, Rajshahi at 76.25 MW, Rangpur at 53.97 MW, Sylhet at 40.62 MW, Mymensingh at 59.22 MW, Chattogram at 76.52 MW, and Cumilla at 75.92 MW (`ablation_raw.json`, variant A6). Dhaka MAE exceeds the next-largest division (Khulna) by more than 205 MW and accounts for the dominant share of the macro numerator despite equal per-division weighting in the macro average.

The same concentration pattern appears for the historical hybrid reference B07 in Table S1 (Dhaka MAE 299.78 MW versus peripheral values generally below 90 MW), indicating that high absolute error in the capital division is a persistent feature of the test period across PF-STGT configurations rather than an artefact isolated to S2. S2 reduces Dhaka MAE by 5.80 MW relative to B07 (293.98 versus 299.78 MW) while retaining a ninefold gap between Dhaka and Barishal MAE. The interpretation is that remaining demand error is structurally concentrated in the largest load centre; the limitation is that per-region signed residuals and box-plot distributions for S2 are not published in the frozen supplementary materials, restricting regional diagnostics to absolute MAE.

3) Demand versus stress error comparison: On the common test split, S2 explains approximately 68.4% of macro demand variance (R^2 = 0.684) while explaining 74.5% of OSI variance (R^2 = 0.745). Absolute demand error is reported in megawatts (macro MAE 88.65 MW), whereas OSI error is reported on the unit interval (MAE 0.0371), so the two tasks are not directly comparable in physical units; the relevant comparison is relative explanatory power and absolute error on each task's native scale.

Demand MAPE of 6.55% indicates that typical regional percentage deviations remain modest at the macro level, but megawatt RMSE of 127.29 MW shows that demand errors on high-load days or in high-load divisions can exceed the mean absolute level. OSI MAE below 0.04 with R^2 of 0.745 indicates that graph-level stress forecasts track the held-out OSI series more closely in variance terms than demand forecasts do, despite demand being the primary optimisation weight in training. Representative test-day case records from Experiment 04 illustrate isolated stress deviations: on 2024-09-08 (high-stress stratum), observed OSI is 0.565 while the S2 prediction is 0.429, whereas on 2024-10-27 (typical-demand stratum) observed OSI is 0.426 and the prediction is 0.411 (`case_studies.md`). The benchmark demonstrates that large single-day OSI gaps can occur on the test split, but four published test case studies are insufficient to characterise systematic temporal or regime-dependent stress bias; that mechanism remains unresolved in the frozen artefacts.

4) Coalition-level error structure: Fig. 6 and Fig. 7 present frozen coalition-level summary plots from Experiment 04 for the S2 checkpoint, visualising how forecast deviations associate with grouped input coalitions on the stress task (Fig. 6) and on Dhaka demand (Fig. 7). On the stress task, limitation stack (G8) and calendar/trend (G6) coalitions show the widest spread among the plotted groups; grid aggregates (G7) and national generation scalars (G10) exhibit narrower spread (Fig. 6; frozen Exp04 coalition rankings). On Dhaka demand, calendar/trend (G6), engineered lags and rolling statistics (G4), and national generation scalars (G10) show the widest spread, with regional demand and supply blocks (G1, G2) at intermediate levels (Fig. 7).

The plots indicate that remaining forecast error co-occurs with variability in calendar, limitation, and lagged-demand input coalitions rather than being uniformly distributed across all feature groups. The repository evidence does not establish whether errors are driven by mis-specification of those coalitions or by unmodelled events outside the input tensor; causal attribution is not resolved by the error summaries alone. The limitation is that these coalition plots summarise association structure on the frozen evaluation sample and do not replace per-window residual traces or signed bias tests for S2.

5) Temporal behaviour and residual diagnostics: The predefined error-analysis protocol specified temporal, extreme-event, and graph-conditioned error segments, but no executed temporal MAE bins, monthly residual profiles, or pooled residual store for S2 appear in the frozen results package. Temporal error behaviour on the 2024 test period therefore cannot be reported as a segmented time series from repository outputs.

Experiment 04 includes four representative test dates (2024-10-27, 2024-04-28, 2024-12-19, 2024-09-08) spanning typical-demand, high-demand, low-demand, and high-stress strata (`case_studies.md`). These records show day-specific demand totals between 8,866 MW and 16,418 MW and OSI values between 0.344 and 0.565 on the test split, confirming that the held-out year contains diverse operating points, but they do not constitute a systematic temporal error audit. Equivalent pooled signed-residual statistics are not available for S2 in the frozen evaluation record; the historical PF-STGT reference B07 is therefore discussed below for contextual orientation only. Experiment 02A reports pooled residual diagnostics for B07—mean residual -19.58 MW, median -18.97 MW, residual skew -2.998, and pred-to-actual standard-deviation ratio 0.983 with attenuated peak tracking (`residual_analysis.md`, `prediction_distribution_analysis.md`). Those values characterise the prior hybrid-graph configuration and cannot be transferred to S2 without a dedicated residual audit of the final checkpoint. The limitation is that temporal and signed-residual characterisation of S2 remains incomplete in the present evaluation.

6) Deployment implications: The frozen error evidence supports deploying S2 as a joint forecaster with macro demand MAE of 88.65 MW and OSI R^2 of 0.745, while treating Dhaka as a separately monitored division: its MAE of 293.98 MW dominates national error budgets despite S2 achieving the lowest audited Dhaka MAE among models with published division tables (Section V-B). Operators should expect comparatively well-calibrated graph-level stress forecasts on average, with awareness that high-stress test days documented in Experiment 04 can exhibit OSI under-prediction exceeding the macro stress MAE.

Coalition-level summaries in Figs. 6 and 7 suggest that deployment monitoring should include limitation-stack and calendar inputs alongside lagged demand channels, because forecast deviations cluster with variability in those coalitions on the frozen evaluation. Without executed temporal segmentation under the planned error-analysis framework or S2 signed-residual histograms, routine operational guardrails should rely on published macro and per-region MAE/R^2 thresholds, division-level Dhaka reporting, and periodic stress-case review rather than on repository-supplied uncertainty intervals or automated bias alarms. A limitation of the current evaluation is the absence of complete S2 residual time-series diagnostics and uncertainty bounds; remaining error structure is therefore documented through aggregate metrics, regional concentration, and coalition associations rather than through signed residual traces or forecast confidence intervals on the test split.

F. Explainability Analysis

1) Scope and attribution methods: The explainability evaluation applies post-hoc attribution [18], [20] to the frozen S2 checkpoint (A6, seed 42) without retraining or altering benchmark metrics from Sections V-A–V-E. Table VII consolidates headline outputs from Experiment 04: grouped GradientSHAP attributions (reported as coalition magnitudes —ϕ—), coalition-level permutation importance, exported spatial and temporal attention weights, and 24 stratified case studies (20 validation, 4 test). Coalition —ϕ— values are obtained through a Captum-compatible integrated-gradients loop using a GradientSHAP approximation with 25 steps and a zero baseline—a SHAP-style attribution pipeline rather than classical exact Shapley enumeration (`shap_summary.md`; `shap_engine.py`).

Cross-method checks compare gradient attributions with permutation importance and OSI component decomposition; attributions reflect model behaviour on the validation sample, not physical causation.

2) Coalition-level gradient attribution: On the global stress task, grouped GradientSHAP magnitudes rank limitation stack (G8) first at —ϕ— = 0.0191 and calendar/trend (G6) second at —ϕ— = 0.0190, followed by grid aggregates (G7) at 0.0087 and national generation scalars (G10) at 0.0082 (Table VII; `shap_summary.md`, validation n = 20). For Dhaka demand, calendar/trend (G6) records —ϕ— = 162.34, engineered lags and rolling statistics (G4) record 101.26, and national generation scalars (G10) record 91.44; regional demand and supply blocks (G1, G2) follow at 77.72 and 77.41 respectively, with limitation stack (G8) far lower at 23.15 on this task.

Stress and demand heads emphasise different coalition families: limitation and calendar channels dominate stress attributions; calendar, lag, and national-generation channels dominate Dhaka demand attributions.

3) Spatial node attribution: Fig. 8 presents the node-level importance heatmap combining mean GradientSHAP node mass and spatial attention on the correlation graph. Dhaka receives the highest mean node attribution mass at 340.36, followed by Rajshahi (110.32), Khulna (108.91), and Mymensingh (93.84); peripheral nodes record substantially lower mass, with Rangpur at 1.99 and Chattogram at 6.21 (Table VII; `node_attribution.md`). The Spearman correlation between learned spatial attention and the frozen correlation adjacency is ρ = 0.422.

TABLE III
BENCHMARK COMPARISON (TEST SET).

ID	Model	MAE	RMSE	MAPE	R^2	OSI MAE	OSI R^2
S2	Corr.-Only PF-STGT	88.65	127.29	6.55	0.684	0.0371	0.745
B07	PF-STGT W20 hybrid	93.31	128.81	6.76	0.674	0.0499	0.585
B02	Random Forest	97.03	156.99	7.04	0.984	0.0481	0.555
B03	XGBoost	109.73	178.53	7.99	0.979	0.0497	0.525
B06	T-GCN	257.21	301.06	15.72	−0.483	0.0891	−0.304

Attribution mass concentrates on high-load divisions with strong Dhaka connectivity (ρ = 0.422 between attention and correlation adjacency), consistent with regional error concentration in Section V-E.

4) Temporal attribution: Fig. 9 plots mean temporal attention weights α_t across the seven-day input window. Weights are near-uniform, with the largest mean weight at lag t-6 (α = 0.1622), followed by t-3 (0.1459) and t-0 (0.1435); remaining lags range from 0.1348 to 0.1405 (`temporal_attribution.md`). Case-study records most frequently list t-6 among the top three temporal lags (Table VII).

Temporal attention is near-uniform across the seven-day window (peak at t-6, α = 0.1622), suggesting broad rather than single-lag dependence.

5) OSI dual-path stress attribution: Fig. 10 compares grouped GradientSHAP stress attributions with OSI component drivers (reserve margin c_2 versus limitation stack c_3) across the 24 case studies. The dominant stress coalition in case records is G8 (limitation_stack); peak-demand strata sometimes rank G7 (grid_aggregates) highest. Dual-path agreement between the top gradient-attribution coalition and the OSI component driver occurs in 13 of 24 cases (52.2%) (`stress_attribution.md`; `case_studies.md`). Stress explanations should report this partial alignment and the 47.8% disagreement rate.

6) Cross-method validation and limitations: Permutation importance partially corroborates stress gradient-attribution rankings (Spearman ρ = 0.366) but disagrees with Dhaka demand gradient-attribution order (ρ = -0.564) (Table VII; `feature_importance.md`). On demand, permutation ranks regional supply block (G2) highest by mean MAE degradation (1.96 MW) while GradientSHAP ranks calendar/trend (G6) highest (—ϕ— = 162.34); engineered lags (G4) show high —ϕ— magnitude (101.26) but low permutation degradation (0.013 MW). On stress, limitation stack (G8) leads both gradient-attribution and permutation tables, though permutation deltas are numerically small (top stress permutation $\Delta \approx 3.8 \times 10^{-5}$).

Gradient and permutation rankings disagree on demand (ρ = -0.564) but partially agree on stress (ρ = 0.366). Attributions (n = 20 validation windows; 24 case studies) describe model behaviour, not causal validation; explanations should combine Figs. 8–10 with explicit cross-method discordance reporting.

TABLE IV
BENCHMARK WILCOXON TESTS (DEMAND MAE; REF. B07).

Comparison	Median ΔMAE	p (two-sided)	Bonferroni sig.
B07 vs B01	-58.98	1.72×10^{-31}	Yes
B07 vs B02	-4.92	0.00135	Yes
B07 vs B06	-160.66	1.48×10^{-40}	Yes

TABLE V
ABLATION STUDY RESULTS (TEST SET).

ID	Variant	MAE	R^2	OSI MAE	OSI R^2
A4	Single-Task	86.89	0.731	–	–
A6 (S2)	Corr. Graph	88.65	0.684	0.0371	0.745
A1	PF-STGT W20	93.31	0.674	0.0499	0.585
A5	Geo Graph	97.98	0.554	0.0340	0.764

TABLE VI
ARCHITECTURE COMPARISON (S1–S4).

ID	Model	MAE	R^2	OSI R^2
S2	Corr.-Only PF-STGT	88.65	0.684	0.745
S3	No-Transformer	92.64	0.671	0.701
S1	PF-STGT W20	93.31	0.674	0.585
S4	Corr. + No-Trans.	114.63	0.362	0.747

TABLE VII
EXPLAINABILITY SUMMARY (S2).

Metric	Value
Top stress coalition (G8 $\lvert\phi\rvert$)	0.0191
Top Dhaka demand coalition (G6 $\lvert\phi\rvert$)	162.34
Attention–adjacency Spearman ρ	0.422
SHAP–permutation ρ (demand)	-0.564
OSI driver agreement (cases)	52.2%

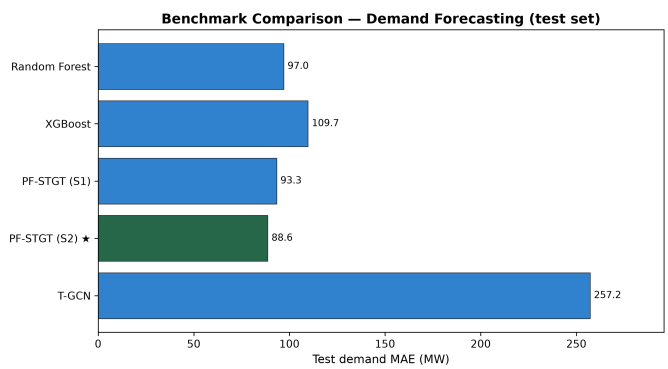

Fig. 4. Test-set macro demand MAE benchmark comparison.

VI. DISCUSSION

A. Principal Findings

1) Overall contribution: The completed study delivers a frozen Correlation-Aware Multi-Task Forecasting Framework that jointly forecasts regional demand and graph-level OSI from a single spatio-temporal graph-transformer [7], [17] on Bangladesh national grid data [1]. The principal finding is that one architecture can supply operationally coupled day-ahead load and stress visibility from shared representations. S2 was selected through evidence spanning benchmarks, ablations, architecture selection, error characterisation, and attribution—not design intent alone.

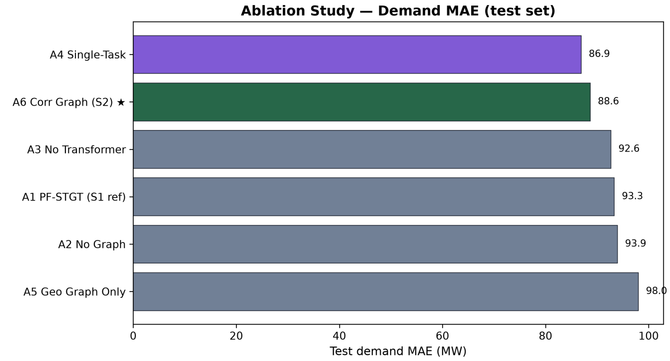

Fig. 5. Ablation study demand MAE on the test set.

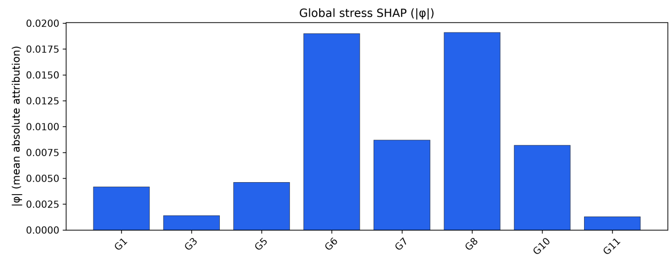

Fig. 6. Grouped SHAP attributions for stress (global).

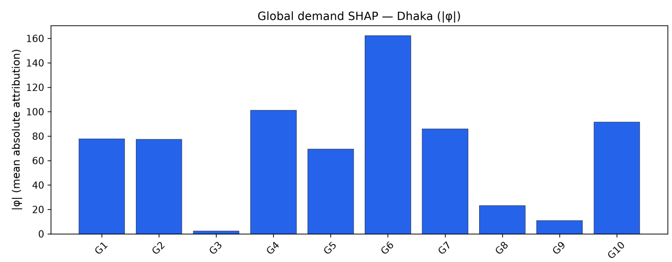

Fig. 7. Grouped SHAP attributions for Dhaka demand.

2) Interaction between demand forecasting and stress prediction: Demand and OSI are complementary rather than redundant within one model. Shared encoding captures calendar, limitation, and grid-context structure; task-specific heads preserve distinct output semantics. Multi-task training [14], [19] under the repaired W20 protocol restores stress learnability that earlier configurations lacked. Lower demand error does not automatically imply better stress fidelity—geographically specialised and demand-only variants illustrate a genuine demand–stress trade-off space.

3) Architecture selection and empirical simplification: Evidence-driven simplification of the graph prior improves the frozen model: correlation-only adjacency outperforms geographical-only and hybrid designs, while geographical-only conditioning degrades demand accuracy. Transformer capacity appears critical on correlation graphs—removal on the correlation topology sharply degrades demand performance—

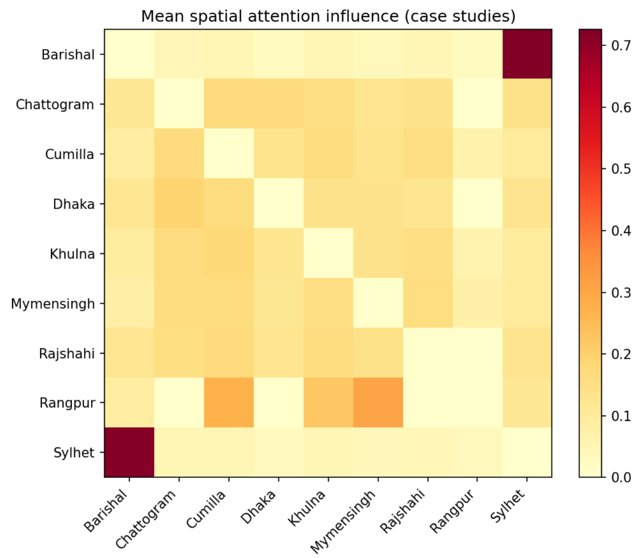

Fig. 8. Node-level attribution heatmap.

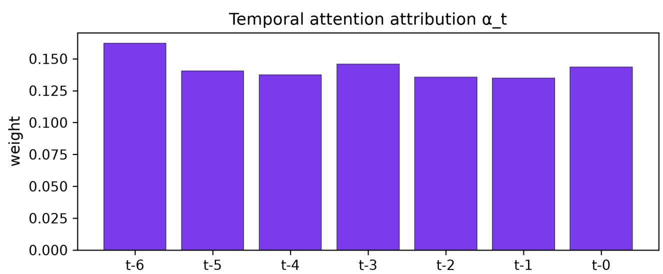

Fig. 9. Temporal attention weights across the lookback window.

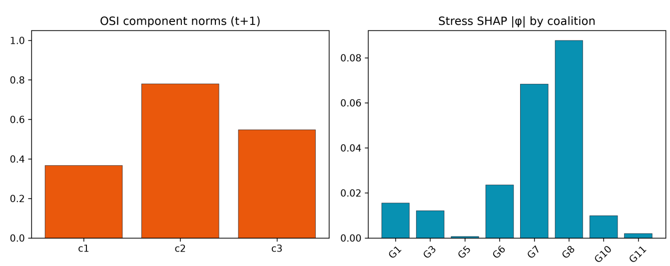

Fig. 10. Dual-path stress attribution vs. OSI component drivers.

even though transformer removal on the hybrid reference is statistically indistinguishable on demand error. Predictive inter-region coupling is better encoded by train-estimated correlation than by static geography alone.

4) Regional robustness and error structure: Regional analysis reveals a robust but heterogeneous forecaster: peripheral divisions achieve low absolute error, while the national load centre dominates the megawatt error budget across model families. S2 improves the capital division relative to earlier PF-STGT and classical references without eliminating structural error concentration in the largest division. Regional robustness here means stable multi-division service with predictable error geography, not uniform accuracy. Error concentration in the highest-variability division persists across architectures, suggesting remaining demand difficulty reflects scale and volatility of the dominant node rather than graph conditioning alone. Operators should report Dhaka separately in deployment monitoring while relying on the model for consistent peripheral and mid-scale service.

5) Explainability contribution: Post-hoc attribution on S2 shows interpretable coalition families—limitation and calendar for OSI, calendar and lag channels for the principal load division—with spatial mass on high-load nodes and moderate attention–adjacency alignment. Dual-path comparison achieves partial agreement (52.2% of case studies); demand attributions disagree with permutation rankings (ρ = -0.564). Attributions offer transparency, not causal validation: limitation-stack and calendar coalitions should headline stress dashboards; demand views should report gradient-based and permutation-based rankings separately.

6) Scientific significance: Correlation-graph PF-STGT multi-task learning jointly models regional demand and operational stress on a chronologically held-out Bangladesh year, with statistically supported demand improvement over the historical hybrid reference and stronger stress R^2 within the frozen suite. The evidence-locked chain—task repair, benchmarks, ablation, architecture freeze, error geography, and attribution—is reproducible without rerunning training. The work extends spatio-temporal forecasting beyond load-only benchmarks to coupled stress indices and shows that graph priors and multi-task objectives require empirical tuning. Day-ahead joint load-and-stress forecasting is feasible as a single-model service when graph structure, task weighting, and division-level monitoring co-design with the evaluation evidence (Section V).

B. Comparison with Existing Literature

1) Graph-based spatial forecasting: The present study finds that train-estimated correlation adjacency outperforms geographical-only and hybrid geographical–correlation priors for joint regional demand and OSI forecasting on the Bangladesh national grid, while a shallow spatio-temporal graph-convolution baseline demonstrates substantially lower predictive performance than the graph-transformer family in the frozen benchmark suite. Prior graph-neural work in power systems—represented in the reviewed corpus by a small set of high-relevance spatio-temporal GNN and graph-attention studies [8], [9], [26]—typically couples nodes through geographic proximity, charging-network topology, or integrated-energy-system structure on US and European case studies rather than on developing-economy division-level operational records.

Graph designs validated on residential or microgrid networks do not transfer automatically to a nine-division national system with strong inter-region coupling and nationally co-visible limitation dynamics [15], [16], [25]. Data-driven correlation graphs can encode predictive spatial structure where static geography is insufficient; external replication on other South Asian national grids remains an open gap.

2) Transformer-based temporal modelling: The frozen architecture-selection record shows that temporal transformer capacity appears critical when the graph prior is correlation-based, whereas transformer removal on the historical hybrid reference is not statistically distinguishable on demand error—a pattern not commonly reported in transformer forecasting papers that evaluate a single backbone on a fixed graph. Transformer-based load forecasting [22], [23] constitutes a larger but still minority share of the reviewed literature, often applied to residential sequences, EV charging, or multivariate energy series without multi-node national stress heads.

Transformer contribution is conditional on graph topology in this multi-task setting; near-uniform temporal attention suggests limited lag selectivity. Transformer value in spatio-temporal energy forecasting should be reported with graph context and ablation evidence, not assumed from single-configuration benchmarks.

3) Multi-task demand and operational stress learning: A principal outcome of this study is that regional demand and graph-level OSI can be learned jointly from shared representations when task weighting is explicitly repaired, yielding a favourable joint configuration relative to single-task and stress-weak baselines, while still exhibiting a documented demand–stress trade-off against a demand-only upper bound. The reviewed literature rarely couples continuous multi-node demand forecasting with operational-stress-style targets in one model: multi-task entries in the corpus are sparse, and the shedding cluster predominantly addresses control, optimisation, or under-frequency response rather than day-ahead forecasting of coupled load and pressure indices.

Relative to hierarchical multi-energy proposals [19], [24], this formulation couples limitation-aware operational stress with evening-peak demand rather than sectoral energy vectors. Multi-task learning is viable when tasks are operationally related yet non-redundant, but not as a default regulariser without loss calibration; cross-dataset task-interference evidence remains limited.

4) Explainability in graph-temporal forecasting: Post-hoc attribution on the frozen model shows coherent coalition-level and spatial patterns—limitation and calendar structure for stress, calendar and lag channels for the principal load division—with partial dual-path agreement against OSI component drivers and discordance between gradient-based and permutation-based demand rankings. Explainability appears in only a small fraction of the reviewed corpus; where present, it often serves microgrid shedding control, distillation, or generic interpretable ML frameworks rather than multi-node graph-transformer forecasts with operator-facing dual-task outputs.

Compared with SHAP-centric shedding studies [11], [18] and XAI reviews [12], this work embeds multi-method explainability in a national graph-transformer pipeline [13] and reports method disagreement explicitly. Prospective operator validation of attribution outputs remains an open gap.

5) Bangladesh smart-grid context and evaluation discipline: The Bangladesh case [1] and spatial load-shedding literature [10] provide contextual anchors for interpreting the frozen evaluation record.

The programme addresses a geographic void: no reviewed comparator used Bangladesh division-level data, and developing-economy national grids remain under-represented in graph-transformer benchmarks. The frozen evaluation chain—chronological hold-out, benchmark and ablation suites, inferential testing, and locked artefacts—provides reproducible internal evidence without claiming superiority over unbenchmarked external systems. External validity requires new datasets and independent replication beyond the Bangladesh timeline.

C. Practical Implications

1) Day-ahead operational planning: S2 delivers the strongest joint demand-and-stress credentials on the held-out test year, supporting a single service that updates regional evening-peak load and national OSI from one input window. Planners could review paired forecasts on the same daily cycle as the one-day horizon and seven-day lookback. Deployment requires the frozen S2 checkpoint, feature and graph artefacts, and MD5-locked preprocessing rules. Forecast accuracy gains (Section V) are planning diagnostics only—not demonstrated reductions in operational risk or economic cost.

2) Regional monitoring: Error concentrates in the national load centre while peripheral divisions show low megawatt deviations; macro summaries may understate planning sensitivity in the largest division. Monitoring should treat the capital division as a separate diagnostic alongside macro dashboards, with two-tier reporting (national summary plus division panels for Dhaka, Khulna, and high-residual nodes). Per-region signed-residual diagnostics for S2 are incomplete; protocols should track division MAE and audited R^2 where available.

3) Joint demand and OSI forecasting: Demand-only and stress-optimal configurations occupy a documented trade-off space; S2 is the empirically preferred joint option, not simultaneous optimum on every task. Organisations needing load only may consider a demand-only variant; centres requiring synchronised load and stress visibility may favour S2. OSI claims rest on point estimates without stress-specific hypothesis tests; joint monitoring should treat OSI as a complementary indicator, not an automatic control trigger.

4) Explainability for operator support: Limitation-stack and calendar coalitions dominate stress explanations; calendar, lag, and national-generation coalitions feature for the principal load division. Stress views should foreground limitation and calendar channels; demand views should present gradient-based and permutation-based attributions separately and flag dual-path disagreement strata. Attributions are model-local and must not substitute for engineering judgement on high-stress days.

5) Deployment considerations: The frozen programme provides a reproducible evidence base for considering S2 but is not a field deployment trial. Integration requires the chronological feature store, correlation adjacency, W20 checkpoint, and inference pipeline from the experimental setup. Shadow-mode operation alongside incumbent tools is recommended before cutover. Monitoring should track limitation and calendar inputs and separate Dhaka reporting; absence of S2 residual time-series diagnostics and uncertainty bounds limits automated bias alarms.

D. Limitations

1) Dataset scope: The empirical evidence is bounded by a single chronologically split daily record of the Bangladesh national power system: nine administrative divisions, a seven-day input window, a one-day evening-peak forecast horizon, and a feature regime comprising nine node-level and seventeen global channels derived from the frozen preprocessing pipeline. That design supports reproducible within-dataset comparison but does not span alternative temporal resolutions, intra-day dispatch horizons, or feature sets outside the locked parquet artefacts.

Model selection, graph construction, and OSI definition were calibrated on one national timeline (train through mid-2023, validation through early 2024, test on 264 windows through late 2024) with MD5-locked inputs. Conclusions apply to this feature geometry and calendar span, not to arbitrary telemetry schemas or revised stress formulations without re-execution.

2) Bangladesh-only evaluation: All reported benchmarks, ablations, error characterisations, and explainability outputs derive from the Bangladesh national grid case alone; no independent replication on other South Asian or developing-economy systems is included in the frozen record. The study addresses a geographic void in the reviewed literature but remains a single-country evidence base.

Correlation-graph priors and limitation-stack coalitions were estimated from Bangladesh records and may not transfer where coupling or stress semantics differ. External validity is limited to systems with comparable divisional structure and daily planning cadence.

3) Benchmark scope: Comparative evidence is confined to the frozen internal suite—seven reference models (B01–B07) plus the final proposed configuration S2—and six ablation variants (A1–A6) drawn from the same PF-STGT family and training protocol. No external state-of-the-art leaderboard, proprietary operator model, or real-time control benchmark participates in the reported rankings.

Inferential testing applies primarily to macro demand MAE; direct S2-versus-random-forest testing is absent. Classical ensembles show favourable macro R^2 without leading on megawatt MAE; demand-only A4 attains lower demand error than S2, documenting an explicit demand–stress trade-off. "Leading" performance is bounded to the frozen suite, not unbenchmarked external systems.

4) Uncertainty estimation: The evaluation reports point forecasts and aggregate error metrics—MAE, RMSE, R^2, and MAPE on demand; MAE and R^2 on OSI—without prediction intervals, ensemble spread, conformal bounds, or other uncertainty quantification on the held-out test split. Bootstrap confidence intervals exist for selected paired demand MAE contrasts but do not cover OSI residuals or per-division S2 forecasts.

Stress-specific hypothesis tests, OSI confidence intervals, and S2 pooled signed-residual diagnostics were not executed; temporal error segmentation remains unreported. OSI gains rest on point estimates (Section V); demand improvements retain inferential backing.

5) Explainability limits: Post-hoc attribution was applied to the frozen S2 checkpoint using grouped integrated gradients, permutation importance, attention export, and stratified case studies; these methods describe model behaviour on the evaluation sample and do not establish physical causation or operator-ground-truth driver validation. Global gradient attributions for demand were computed on twenty validation windows; dual-path OSI comparison draws on twenty-four stratified days, of which only four lie on the test split.

Cross-method agreement is partial: OSI component drivers align with the top gradient-attribution coalition in 52.2% of audited case studies, demand gradient and permutation rankings disagree (Spearman ρ = -0.564), and spatial attention correlates moderately with the correlation adjacency (ρ = 0.422). Near-uniform temporal attention weights further limit claims of sharp lag selectivity from attribution outputs alone. Attributions describe input sensitivities only; they do not validate OSI physics or substitute for engineering judgement (Table VII).

6) Deployment limits: The completed programme is an offline, chronologically held-out evaluation on locked artefacts; it does not include field deployment, operator-in-the-loop trials, shadow-mode operation, or quantification of downstream dispatch, reserve, shedding, or economic outcomes. Production integration would depend on reproducing the frozen feature store, correlation adjacency, W20 multi-task checkpoint, and inference pipeline under the same train-only preprocessing rules that governed reported scores.

Single-seed training (seed 42), incomplete S2 residual stores, and absent uncertainty bounds limit operational readiness assessment to published accuracy diagnostics. Practical implications (Section VI-C) are advisory recommendations, not measured deployment outcomes.

E. Future Research Directions

1) External validation: The completed study establishes that correlation-graph PF-STGT multi-task learning can jointly forecast regional evening-peak demand and a graph-level Operational Stress Index on a chronologically held-out year of Bangladesh division-level data, with statistically supported demand improvement over the historical hybrid reference within the frozen benchmark suite. That evidence is internally rigorous but geographically singular.

Future studies could replicate the frozen protocol—chronological splitting, joint demand–stress targets, graph ablation, and multi-method attribution—on comparable South Asian national systems. External validation would separate architecture-specific gains from country-specific feature geometry.

2) Richer graph construction: Ablation evidence shows that correlation-only adjacency outperforms geographical-only and hybrid priors for joint demand and stress performance on this dataset, while a geographical graph variant attains superior OSI point estimates at materially higher demand error, documenting a demand–stress trade-off rather than a single optimal topology. Transformer capacity further appears conditional on graph choice: removal on the correlation graph degrades demand performance sharply, whereas hybrid-topology simplification does not yield statistically distinguishable demand change.

Future work could evaluate adaptive correlation graphs, multi-graph attention, and physics-informed edge constraints under the same multi-task objective. This would clarify whether correlation-graph simplification is a robust national-grid principle or a Bangladesh-specific optimum.

3) Uncertainty-aware forecasting: Demand-side inferential support in the frozen record includes paired Wilcoxon tests and bootstrap confidence intervals on macro MAE contrasts, whereas OSI performance is reported through point estimates without stress-specific hypothesis tests, prediction intervals, or pooled signed-residual diagnostics for the final checkpoint. Planners reviewing joint forecasts therefore lack distributional statements on per-division error and stress uncertainty on the held-out split.

Probabilistic extensions—ensembles, quantile heads, or conformal calibration—could supply interval-valued demand and OSI outputs. Executing predefined residual and temporal error protocols on S2 would enable signed-bias characterisation and OSI inferential testing.

4) Explainability validation: Post-hoc attribution on the frozen model surfaces coherent coalition-level and spatial patterns—limitation and calendar structure for stress, calendar and lag channels for the principal load division—with explicit documentation of partial dual-path agreement against OSI component drivers and discordance between gradient-based and permutation-based demand rankings. Transparency is demonstrated on frozen artefacts rather than through prospective validation with domain experts.

Future studies could expand attribution coverage on held-out test windows, standardise dual-task reporting templates, and conduct controlled operator studies measuring whether discordant explanation views affect forecast trust.

5) Deployment studies: The frozen evaluation programme delivers reproducible offline accuracy diagnostics, architecture selection, and post-hoc analysis on locked data artefacts, but it does not measure runtime integration, shadow-mode operation, data-drift response, or downstream planning outcomes. Practical implications in Section VI-C therefore remain conditional on evidence not collected in the present study.

Future studies could implement shadow-mode forecasting, document retraining triggers, and record operator review of paired demand–OSI dashboards under realistic planning cycles, connecting offline accuracy to control-centre behaviour.

VII. Conclusion

Day-ahead planning in developing-economy national grids requires synchronised visibility over regional load and system-wide operational pressure, yet forecasting practice often treats demand prediction and stress monitoring as separate pipelines. This study investigated whether a single spatio-temporal framework can jointly forecast next-day regional evening-peak demand and a graph-level Operational Stress Index from shared historical records on the Bangladesh national power system [1], under chronological evaluation with classical, recurrent, and graph-convolution baselines and systematic ablation of graph priors, architectural components, and task formulation.

The Correlation-Aware Multi-Task Forecasting Framework implements Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT) learning: a seven-day lookback of node-level and national context features over nine divisions, train-estimated correlation-graph conditioning, and parallel regional demand and system-wide stress outputs through shared spatio-temporal encoding and task-specific heads, with a repaired multi-task protocol that stabilises joint learning. The final frozen configuration—correlation-graph PF-STGT with full temporal capacity and multi-task heads—was selected through an evidence-locked programme spanning benchmark comparison, configuration ablation, architecture simplification, error geography, and post-hoc attribution.

Five substantive outcomes emerge. First, graph-transformer multi-task learning delivers operationally coupled day-ahead load and stress forecasts from one checkpoint on a chronologically held-out national timeline. Second, data-driven correlation adjacency outperforms geographical or hybrid graph priors on this dataset, while temporal encoding remains consequential once the correlation prior is adopted. Third, a genuine demand–stress trade-off exists: demand-only training attains a lower megawatt error ceiling, whereas multi-task training enables informative stress outputs at a modest demand offset, positioning the frozen model as the preferred joint optimum. Fourth, regional diagnostics reveal stable error geography—peripheral divisions are comparatively well served while the national load centre dominates absolute residual mass. Fifth, multi-method post-hoc attribution supplies coalition-level, spatial, and temporal transparency with explicit reporting of partial cross-method and dual-path discordance.

Scientifically, the study contributes a reproducible modelling chain on a previously unreported Bangladesh division-level smart-grid case [4], [12], demonstrating that correlation-graph PF-STGT multi-task learning is viable where limitation dynamics and inter-regional load coupling coexist in operational data. Graph priors, task weighting, and architectural simplification require empirical tuning rather than domain intuition; the contribution is bounded to the evaluated national context and frozen internal benchmark suite.

For power-system practice, a single forecasting service may supply paired regional demand trajectories and a national stress scalar for day-ahead review from the same input window. Operators should treat macro summaries as routine briefings, report the principal load centre separately, review limitation and calendar channels highlighted in error and attribution analyses, and present gradient-based and permutation-based explanation views distinctly where methods disagree. These implications are planning diagnostics grounded in offline accuracy and transparency evidence—not demonstrated improvements in grid reliability, economic outcome, or deployment success.

In sum, this work delivers an empirically frozen framework that jointly models regional demand and operational stress on Bangladesh national grid data through correlation-graph PF-STGT learning, supported by a transparent evaluation record from task formulation through architecture selection and post-hoc analysis. Coupled day-ahead load-and-stress forecasting is achievable as a single-model service when graph structure, multi-task calibration, division-level monitoring, and explanation reporting are co-designed with the collected evidence—appropriately qualified as a completed offline evaluation rather than an operational field trial.

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APPENDIX A
SUPPLEMENTARY MATERIALS

This appendix consolidates supplementary technical material for the main manuscript. All entries reference artefacts under publication freeze `publication-freeze-2026-06-25` (commit `dda83f1d9201d55ad8daf6b4cc0456569a84b6aa`). Table values are copied from frozen CSV/JSON outputs; no experiments were rerun.

A. Complete benchmark tables

1) External benchmark comparison (test set): Macro demand metrics average over nine divisions. **S2** is the final model (Exp. 03, A6); **B07** is the historical PF-STGT W20 hybrid reference (S1). Stress columns apply only to OSI-emitting models.

ID	Model	Demand MAE (MW)	Demand RMSE (MW)	Demand MAPE (%)	Demand R^2	Stress MAE	Stress R^2
S2	Correlation-Only PF-STGT (final)	88.65	127.29	6.55	0.684	0.0371	0.745
B07	PF-STGT W20 hybrid (S1 ref.)	93.31	128.81	6.76	0.674	0.0499	0.585
B02	Random Forest	97.03	156.99	7.04	0.984	0.0481	0.555
B03	XGBoost	109.73	178.53	7.99	0.979	0.0497	0.525
B01	Linear Regression	247.79	597.01	17.32	0.770	0.1074	-1.824
B04	LSTM	237.03	278.67	14.35	-0.242	0.0861	-0.191
B05	GRU	233.48	274.39	14.13	-0.201	0.0863	-0.214
B06	T-GCN	257.21	301.06	15.72	-0.483	0.0891	-0.304

Source: experiments/experiment_02_benchmark_models/benchmark_results.csv (B01–B07); experiments/experiment_03_ablation_studies/ablation_results.csv (S2 = A6).

2) Ablation study (test set): Reference: **A1** = S1 (hybrid graph, multi-task W20); **A6 = S2** (final architecture).

ID	Variant	Graph	Multi-task	Demand MAE (MW)	Demand R^2	Stress MAE	Stress R^2
A4	Single-Task	hybrid	No	86.89	0.731	—	—
A6	**Correlation Graph Only (S2)**	corr	Yes	**88.65**	**0.684**	**0.0371**	**0.745**
A3	No Transformer	hybrid	Yes	92.64	0.671	0.0405	0.701
A1	PF-STGT W20 (S1 ref.)	hybrid	Yes	93.31	0.674	0.0499	0.585
A2	No Graph	hybrid	Yes	93.93	0.701	0.0405	0.701
A5	Geographical Graph Only	geo	Yes	97.98	0.554	0.0340	0.764

Source: experiments/experiment_03_ablation_studies/ablation_results.csv, `ablation_raw.json`.

3) Architecture simplification (test set):

ID	Model	Graph	Transformer	Demand MAE (MW)	Demand R^2	Stress R^2	Active params
S2	Correlation-Only PF-STGT	corr	Yes	88.65	0.684	0.745	749,058
S3	No-Transformer PF-STGT	hybrid	No	92.64	0.671	0.701	451,202
S1	PF-STGT W20 (ref.)	hybrid	Yes	93.31	0.674	0.585	749,058
S4	Corr + No-Transformer	corr	No	114.63	0.362	0.747	451,202

Source: experiments/experiment_03B_architecture_simplification/simplification_results.csv.

4) Per-region demand MAE — S2 (test set): Per-division MAE for S2 is in the frozen ablation record; per-division R^2 is not tabulated (macro demand R^2 only).

Region	S2 demand MAE (MW)
Barishal	32.73
Chattogram	76.52
Cumilla	75.92
Dhaka	293.98
Khulna	88.62
Mymensingh	59.22
Rajshahi	76.25
Rangpur	53.97
Sylhet	40.62

Source: experiments/experiment_03_ablation_studies/ablation_raw.json (variant A6, `per_region_mae`).

5) Per-region verification — B07 and B02 (test set): Division-level MAE and R^2 for B07 and B02 were recomputed in Experiment 02A.

Region	B07 MAE (MW)	B07 R^2	B02 MAE (MW)	B02 R^2
Barishal	30.85	0.4808	32.07	0.6936
Chattogram	75.04	0.6362	79.97	0.5941
Cumilla	71.83	0.7930	73.63	0.7688
Dhaka	299.78	0.5744	311.59	0.6666
Khulna	118.72	0.7353	92.06	0.7835
Mymensingh	58.74	0.7584	85.85	0.6126
Rajshahi	85.70	0.8270	83.13	0.8274
Rangpur	59.77	0.6928	68.41	0.6264
Sylhet	39.35	0.5707	46.54	0.6171

Source: experiments/experiment_02A_classical_benchmark_verification/aggregation_audit.md.

Not in frozen repository: Per-region tables for B01, B04–B06; per-region R^2 for S2; S2 per-window prediction arrays (Exp. 02A stores B07, B02, B03 only).

B. Additional statistical summaries

1) Benchmark Wilcoxon tests (demand MAE): Paired Wilcoxon tests [34] on per-window macro demand MAE (n = 264). Reference: **B07 (S1)**; Bonferroni [35] α = 0.0083 (six comparisons).

Comparison	Median ΔMAE (MW)	p (two-sided)	Cohen's d	Bonferroni sig.	Bootstrap 95% CI
B07 vs B01	-58.98	1.72×10^{-31}	-0.491	Yes	[-193.11, -120.58]
B07 vs B02	-4.92	0.00135	-0.077	Yes	[-8.87, 2.62]
B07 vs B03	-14.13	6.68×10^{-12}	-0.298	Yes	[-22.76, -9.55]
B07 vs B04	-134.42	2.33×10^{-39}	-1.224	Yes	[-158.49, -129.48]
B07 vs B05	-128.94	1.92×10^{-39}	-1.219	Yes	[-154.54, -126.62]
B07 vs B06	-160.66	1.48×10^{-40}	-1.296	Yes	[-179.85, -148.85]

Source: experiments/experiment_02_benchmark_models/statistical_significance.md.

2) Ablation Wilcoxon tests (demand MAE): Paired Wilcoxon tests [34] vs **A1**; Bonferroni [35] α = 0.01 (five comparisons).

Comparison	Median ΔMAE (MW)	p (two-sided)	Bonferroni sig. (worse)	Bootstrap 95% CI
A1 vs A6 (S2)	-5.43	5.5×10^{-5}	No (A6 better)	[-7.17, -2.16]
A1 vs A4	-5.25	0.00284	No (A4 better demand)	[-10.63, -2.40]
A1 vs A5	+3.85	1.48×10^{-4}	Yes (A5 worse demand)	[2.19, 6.90]
A1 vs A2	+2.81	0.301	No	[-3.69, 4.93]
A1 vs A3	-1.13	0.384	No	[-2.35, 1.08]

Source: experiments/experiment_03_ablation_studies/statistical_significance.md.

3) S2 vs S1 and R^2 aggregation note:

Item	Value / note	Source
S2 vs S1 (A6 vs A1)	Median ΔMAE = -5.43 MW; p = 5.5×10^{-5}; bootstrap 95% CI [-7.17, -2.16]	Exp03 statistical summary
B02/B03 reported R^2 in Exp02	Equals **pooled** R^2, not macro R^2	`aggregation_audit.md`
B07 reported R^2	Equals **macro** R^2 (mean of nine per-region R^2)	`metric_verification.md`
Unified macro R^2 (recomputed)	B02: 0.6878; B07: 0.6743; B03: 0.5902	Exp02A

Not in frozen repository: S2 vs B02 Wilcoxon test; OSI hypothesis tests and bootstrap CIs; S2 pooled signed-residual store; Phase 14 temporal error-segmentation outputs.

C. Hyperparameter configuration

Frozen training configuration for S2 (= A6, seed 42), per Experiment 01B W20 repair protocol.

Setting	Value
Architecture	`PFSTGT` + `GraphVariant.CORR` (τ = 0.65)
Tasks	Multi-task: regional demand + graph-level OSI
Loss	L = Huber(demand)/100 + $\lambda_2 \cdot$ MSE(OSI)
λ_2 (stress weight)	20.0
Optimiser	Adam, lr = 5×10^{-4}, weight decay = 10^{-4}
Batch size	32
Max epochs	200 (early stopping epoch 69 for A6)
Early stopping	Patience 15; score = 0.7·(val_demand_MAE/100) + 0.3·val_stress_MAE
Scheduler	ReduceLROnPlateau on validation demand MAE; factor 0.5, patience 5
Gradient clipping	Max norm 1.0
Seed	42
Parameters	749,058
S2 checkpoint	experiments/experiment_03_ablation_studies/checkpoints/A6/seed_42/best.pt
S1 reference checkpoint	experiments/experiment_01B_multitask_optimization_repair/checkpoints/W20/B07/seed_42/best.pt

Supporting documents: experiments/experiment_01B_multitask_optimization_repair/best_configuration.md, experiments/architecture_freeze_revision/final_model_specification.md.

Not in frozen repository: Multi-seed sweeps (seed 42 only); probabilistic head configuration.

D. Feature dictionary

1) Model input tensors:

Tensor	Shape (per batch)	Channels	Description
Node features	(T=7, N=9, F_n=9)	9 per division	Contemporaneous demand/supply/load; lags and rolling mean; regional share and load intensity
Global features	(T=7, F_g=17)	17	Calendar/trend; grid aggregates; limitation stack; weather; generation scalars; shedding indicator

Leakage policy: Same-day OSI(t) is excluded when forecasting OSI(t+1); the engineered stress composite is withheld from global inputs at inference.

2) Node-level channels (F_n = 9 per division):

Channel group	Contents
Power balance (contemporaneous)	Processed evening-peak demand, supply, and load (MW-scale, standardised)
Autoregressive temporal	Demand lag (1 obs.), demand lag (7 obs.), load lag (1 obs.), 7-observation demand rolling mean
Regional derived	Division demand share; load intensity (load / max(demand, ε))

3) Global-level channels (F_g = 17):

Channel group	Contents
Calendar and trend	Day-of-year sin/cos, monotonic trend index, inter-observation gap days, holiday-category ordinal
Grid aggregates	Total regional demand, total regional load, generation reserve
Operational	Total operational limitation (aggregated), any-division shedding indicator (binary)
Weather	Temperature anomaly (month-relative, train-fitted means)
Limitation stack	Gas, coal, water, and maintenance limitation scalars (preprocessed)
National generation	Two preprocessed generation-endpoint scalars

4) Coalition grouping (explainability): Eleven coalitions (Exp. 04 grouped attribution):

ID	Coalition name	Scope	Primary contents
G1	regional_demand_block	Node	Contemporaneous division demand
G2	regional_supply_block	Node	Contemporaneous division supply
G3	regional_load_block	Node	Contemporaneous division load
G4	engineered_lags_rolling	Node	Demand lags and rolling mean
G5	regional_share_intensity	Node	Demand share and load intensity
G6	calendar_trend	Global	Cyclical calendar, trend, gap, holiday summary
G7	grid_aggregates	Global	Total demand, total load, generation reserve
G8	limitation_stack	Global	Fuel, infrastructure, maintenance limitations
G9	weather_anomaly	Global	Temperature anomaly
G10	national_generation_scalars	Global	Generation-endpoint scalars
G11	shedding_indicator	Global	Binary any-division shedding flag

Design registry: results/phases/phase_05A_feature_blueprint/feature_inventory.csv (Phase 05A; matches Section III-D).

Not in frozen repository: Standalone column-name manifest from data/features/*_features.parquet; feature columns are defined in Section III-D and the coalition table above.

E. Graph construction details

1) Final model — correlation graph (S2):

Property	Value
Construction rule	Undirected edge (i,j) retained when train-split Pearson ρ_ij $\geq \tau$ on regional evening-peak demand
Threshold τ	0.65
Train rows for ρ	1,295 daily records (training partition only)
Retained pairs	33 of 36 undirected pairs (91.7% density)
Edge weight	w_ij = ρ_ij (zero diagonal; no self-loops)
Normalisation	Row-normalised adjacency A (outgoing weights sum to unity)
Temporal behaviour	Static: estimated once from training data; fixed for validation and test
Runtime variant	`GraphVariant.CORR` in src/models/pf_stgt.py

Attention bias: Derived from $\mathbf{A}$ (log-scaled positive entries; masked absent edges) for additive spatial masking.

2) Alternative graphs evaluated in ablations:

Variant	Rule	Undirected edges	Density	Notes
Geographical	Border-adjacent divisions; uniform binary weights, row-normalised	21	58.3%	Frozen geographic prior in graphs/adjacency_matrix.csv (MD5-locked)
Hybrid (S1)	Edge if geographic neighbour **or** ρ_ij $\geq$ 0.85; weight = ρ_ij	24	66.7%	Historical reference graph for A1/B07
Correlation (S2)	Edge if ρ_ij $\geq$ 0.65; weight = ρ_ij	33	91.7%	Selected final prior

Design report: graphs/graph_construction_report.md (Phase 08 hybrid rationale); S2 selects correlation-only adjacency per experiments/architecture_freeze_revision/Final_Architecture_Decision.md.

Not in frozen repository: Persisted correlation-adjacency CSV; dynamic or rolling-window graph variants.

F. Additional explainability outputs

Experiment 04 post-hoc analysis on the frozen S2 checkpoint (A6/seed_42/best.pt).

1) Summary reports:

Report	Path
XAI summary	experiments/experiment_04_explainability_analysis/xai_summary.md
SHAP summary	experiments/experiment_04_explainability_analysis/shap_summary.md
Permutation importance	experiments/experiment_04_explainability_analysis/feature_importance.md
Node attribution	experiments/experiment_04_explainability_analysis/node_attribution.md
Temporal attribution	experiments/experiment_04_explainability_analysis/temporal_attribution.md
Stress dual-path attribution	experiments/experiment_04_explainability_analysis/stress_attribution.md
Case studies (24 dates)	experiments/experiment_04_explainability_analysis/case_studies.md
Machine-readable metrics	experiments/experiment_04_explainability_analysis/xai_metrics.json

2) Tabular and figure artefacts:

Label	Content	Path
Table S2	Global grouped SHAP (stress)	results/explainability/shap/global_stress.csv
Table S2	Global grouped SHAP (Dhaka demand)	results/explainability/shap/global_demand_dhaka.csv
Table S3	Permutation ΔMAE (demand)	results/explainability/permutation/demand_importance.csv
Table S3	Permutation Δ (stress)	results/explainability/permutation/stress_importance.csv
Table S4	Case-study records	results/explainability/case_studies/⟨date⟩/ (24 date folders)
Figure S1	Signed stress SHAP bar	experiments/experiment_04_explainability_analysis/figures/figure_shap_bar_stress.png
Figures 3–9	Manuscript explainability figures	manuscript/overleaf/figures/ (copies from Exp04)

3) Cross-method headline metrics (S2):

Metric	Value
Attention–adjacency Spearman ρ	0.422
SHAP–permutation Spearman (demand)	-0.564
SHAP–permutation Spearman (stress)	0.366
OSI driver agreement (case studies)	52.2% (13/24)
Global demand SHAP sample	n = 20 validation windows
Case-study dates on test split	4 of 24

Not in frozen repository: Independent Captum export; operator-in-the-loop records; explainability for ablation variants other than S2.

G. Reproducibility checklist

1) Freeze metadata:

Item	Value
Freeze date	2026-06-25
Designated tag	`publication-freeze-2026-06-25`
Git commit (freeze HEAD)	`dda83f1d9201d55ad8daf6b4cc0456569a84b6aa`
Final model ID	S2 — Correlation-Aware Multi-Task Forecasting Framework
Freeze authority	paper/publication_freeze/Publication_Asset_Freeze.md

2) MD5-locked data artefacts:

File	MD5
data/interim/bangladesh_smartgrid_clean.parquet	`4255024d735a91a4b53b2edee203d0ca`
data/features/train_features.parquet	`b8b3bda95d0fd6cc65f4910d85a98e16`
graphs/adjacency_matrix.csv	`dacb7ac3a827d00a4b61ea9400e75686`

3) Chronological split policy:

Split	Date range	Valid windows (T = 7)
Train	2019-11-21 → 2023-06-15	1,281
Validation	2023-06-16 → 2024-03-19	263
Test	2024-03-20 → 2024-12-30	264

Model selection uses validation only; test is held out until final evaluation. Warm-up skip: seven rows per split.

4) Experiment completeness (frozen):

Experiment	Primary artefact	Status
01 / 01A / 01B	W20 protocol + S1 checkpoint	Complete
02	`benchmark_results.csv`	Complete
02A	Verification reports + B07/B02/B03 predictions	Complete
03	`ablation_results.csv` + A6 checkpoint	Complete
03A / 03B	Investigation reports + simplification CSV	Complete
04	`xai_metrics.json` + figures + CSV tree	Complete

5) Artefacts not included in the frozen repository: The following items are **not** present as executed outputs in the frozen record:

1. Per-region demand R^2 table for S2. 2. Pooled signed-residual diagnostics, residual histograms, or residual time-series store for S2. 3. Materialised per-window prediction arrays for S2 (B07/B02/B03 only in Exp02A). 4. OSI-specific Wilcoxon tests, bootstrap intervals, or multiple-comparison correction on stress metrics. 5. Direct paired inferential test of S2 vs random forest (B02). 6. Temporal error-segmentation bins or monthly residual profiles for S2 (Phase 14 protocol defined but not executed for the final checkpoint). 7. Forecast prediction intervals, ensemble spread, or conformal uncertainty bounds. 8. Persisted correlation-adjacency matrix file (runtime construction from training demand). 9. Multi-seed result tables (seed 42 only). 10. Field deployment, shadow-mode, or operator-in-the-loop trial records.

Reproduce reported results using the checkpoint, MD5 hashes, split policy, and scripts in paper/publication_freeze/frozen_results_inventory.md without modifying frozen artefacts.

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